Exposing SCCM and MSSQL Attack Paths in Hardened Environments

with BloodHound OpenGraph

Chris Thompson

- X: @_Mayyhem
- Slack: @Mayyhem
- GitHub: Mayyhem
- LinkedIn: x24

CHRIS THOMPSON

- Sr. Security Researcher @ SpecterOps
- Red Teamer / Pentester
- Open-source Tool Developer
  - SharpSCCM
  - Misconfiguration Manager
  - Maestro
  - MSSQLHound
- Contributor
  - BloodHound
  - SharpHound
  - AzureHound
  - SCCMHunter
  - SCMKit
  - PXEThiefy
- Trainer/Speaker
  - Black Hat US/EU
  - DEF CON
  - TROOPERS
  - MMS
  - SO-CON
  - SpecterBash
  - IT Defense

X: @_MAYYHEM

SLACK: @MAYYHEM
HTTPS://SLACK.SPECTEROPS.IO

GITHUB: MAYYHEM

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AGENDA

SCCM/MSSQL Attack Paths

- Crash Course
- Attack Demos
- Data Collection
- Exploring the Graph

WHO IS THIS TALK FOR?

- Pentesters
- Red teamers
- Defenders
- IT admins
- IT auditors
- IT leadership
- IT curious

WHAT IS SCCM?

WHAT IS SCCM?

- Allows IT admins to **deploy software** and updates to computers

WHAT IS SCCM?

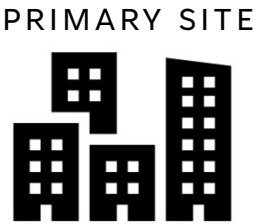
- Released in **1994**
- Currently Microsoft Configuration Manager (ConfigMgr, ConfigMan, MCM)
- Formerly (and still sometimes referred to as):
  - Microsoft Endpoint Configuration Manager (MECM)
  - Microsoft Endpoint Manager Configuration Manager (MEMCM)
  - **System Center Configuration Manager** (SCCM)
  - Systems Management Server (SMS)

WHAT IS SCCM?

- Allows IT admins to **deploy software** and updates to computers
- **Command and control server**

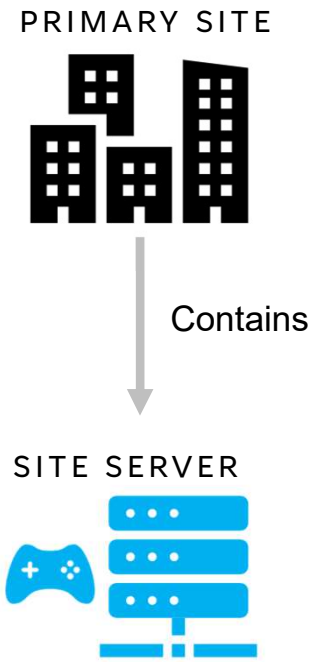
SCCM HIERARCHY

- One installation of SCCM
- Contains one or more **sites**
- This is the **security boundary**



PRIMARY SITES

- ID’d by a 3-character **site code** (e.g., PS1)
- Serve software to managed **client devices**



SITE SERVERS

Orchestrate all actions for:

- the site
- client devices
- site system roles

PRIMARY SITE

SITE SYSTEM ROLES

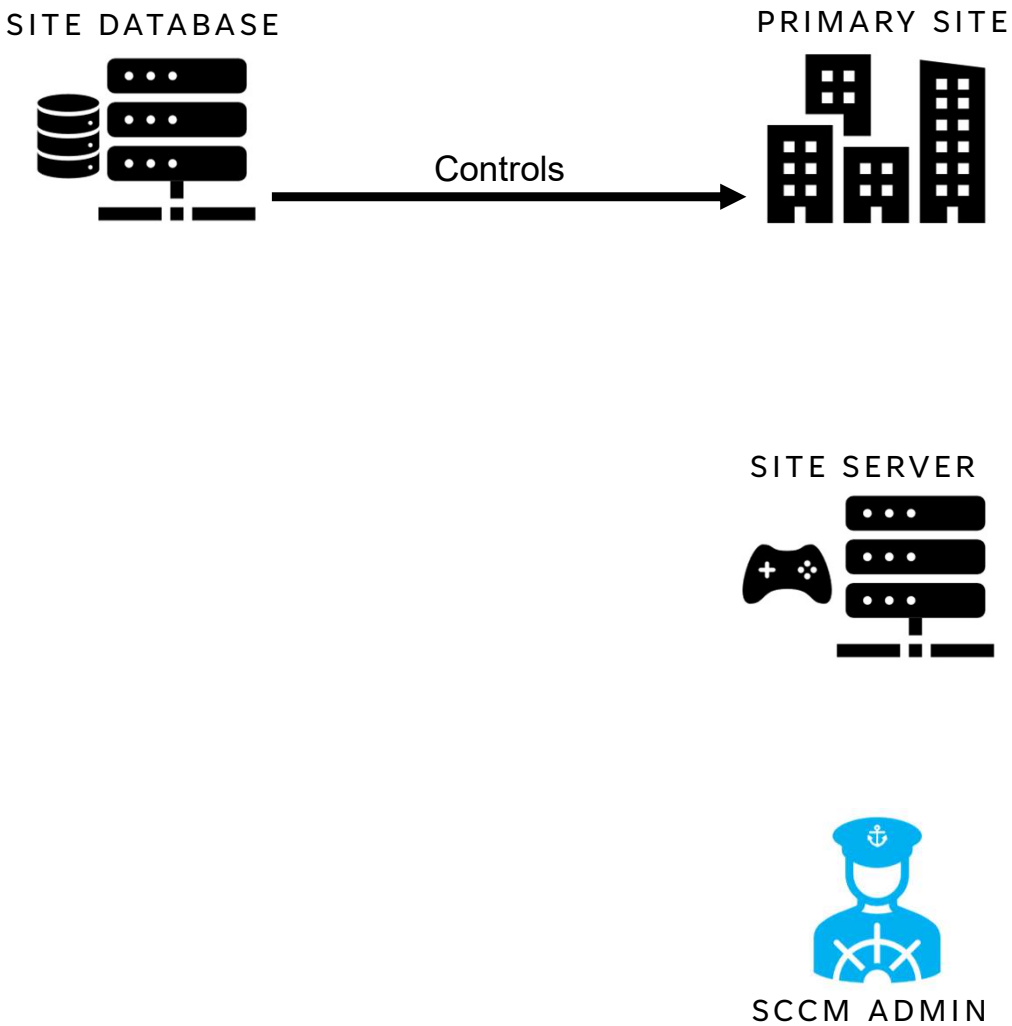
- **Servers** that host one or more services for a primary site
- May be **hosted remotely** from the site server **or on the site server**

SITE DATABASES

- Each site's system of record
- Global configs are **replicated** to all other sites in the hierarchy

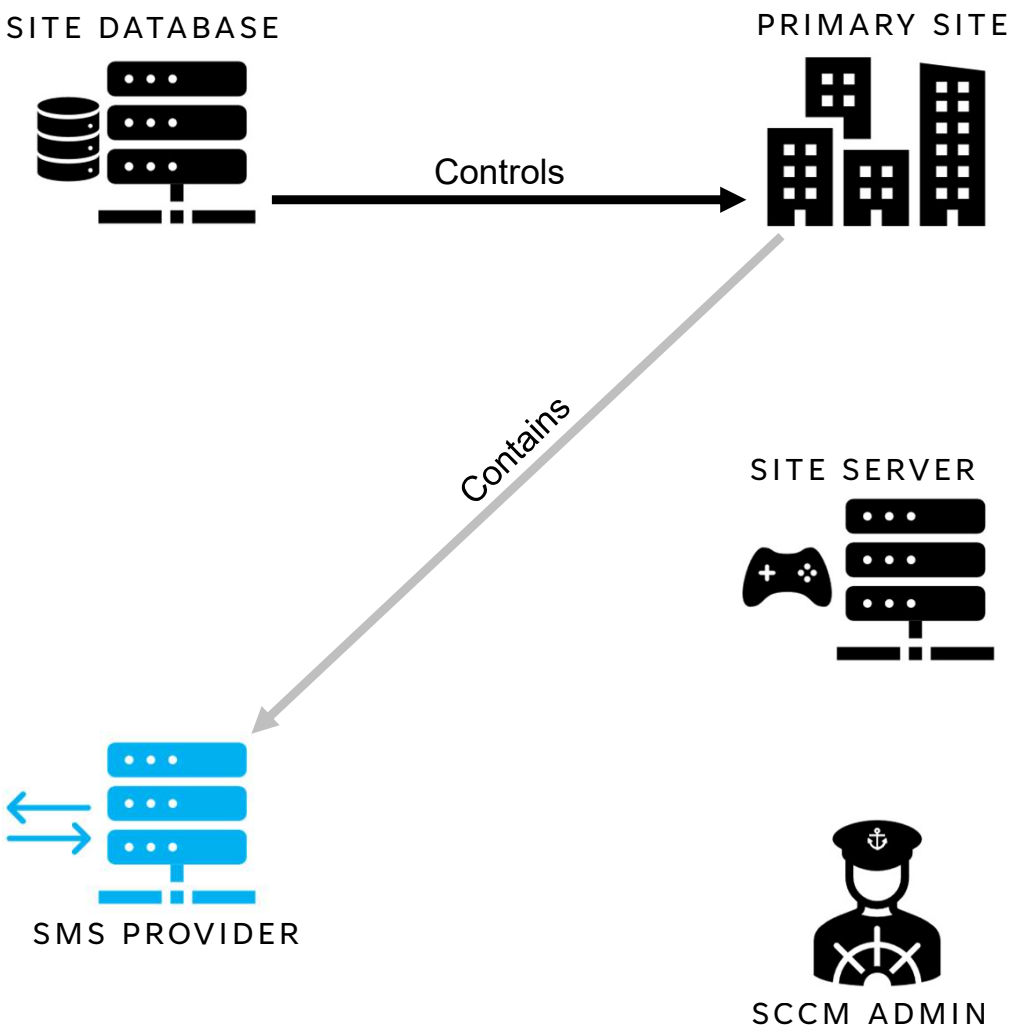
SITE DATABASES

- Each site's system of record
- Global configs are **replicated** to all other sites in the hierarchy
- **Define admin users and roles**



SCCM ADMINS

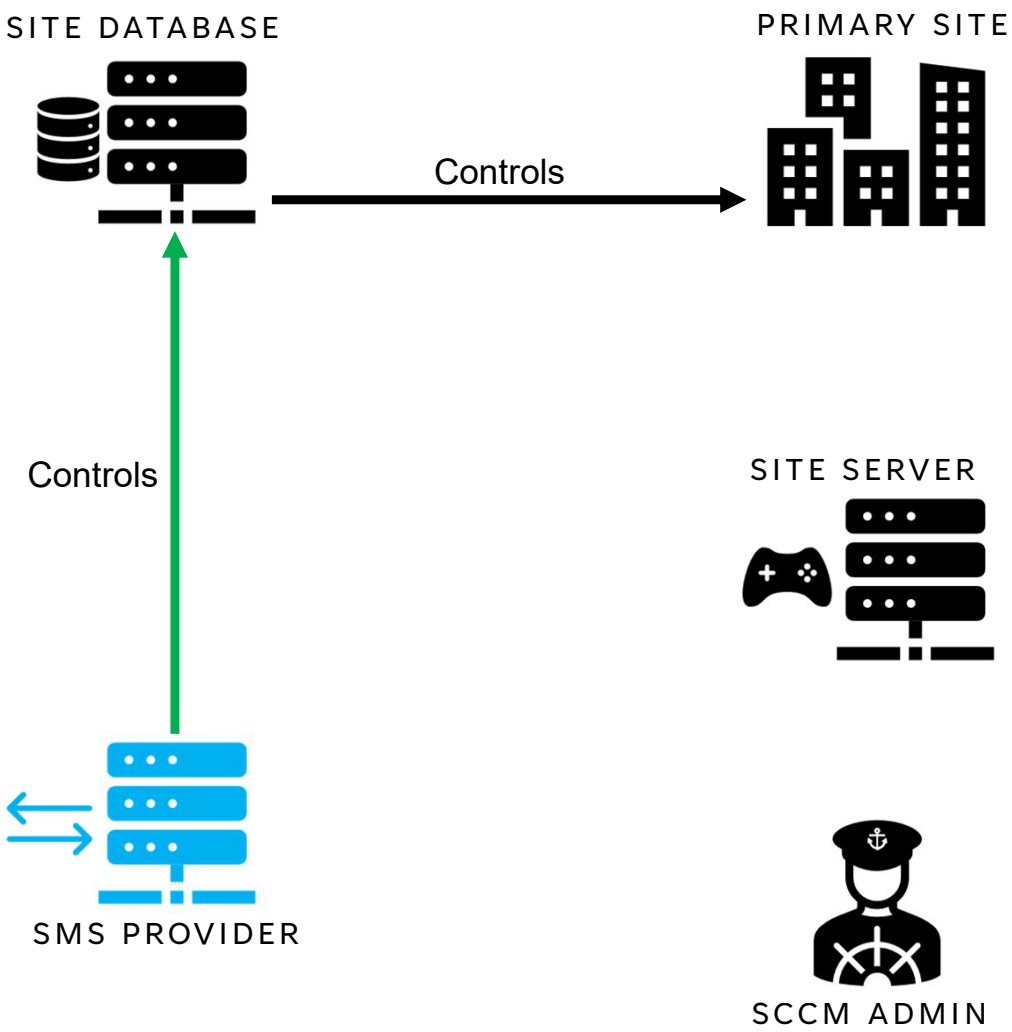
Interact with site database to manage the site... how?



SMS PROVIDERS

Host two APIs for SCCM admins to manage site database:

- **AdminService**
  - REST API (HTTPS)
- **WMI**
  - Windows Management Instrumentation



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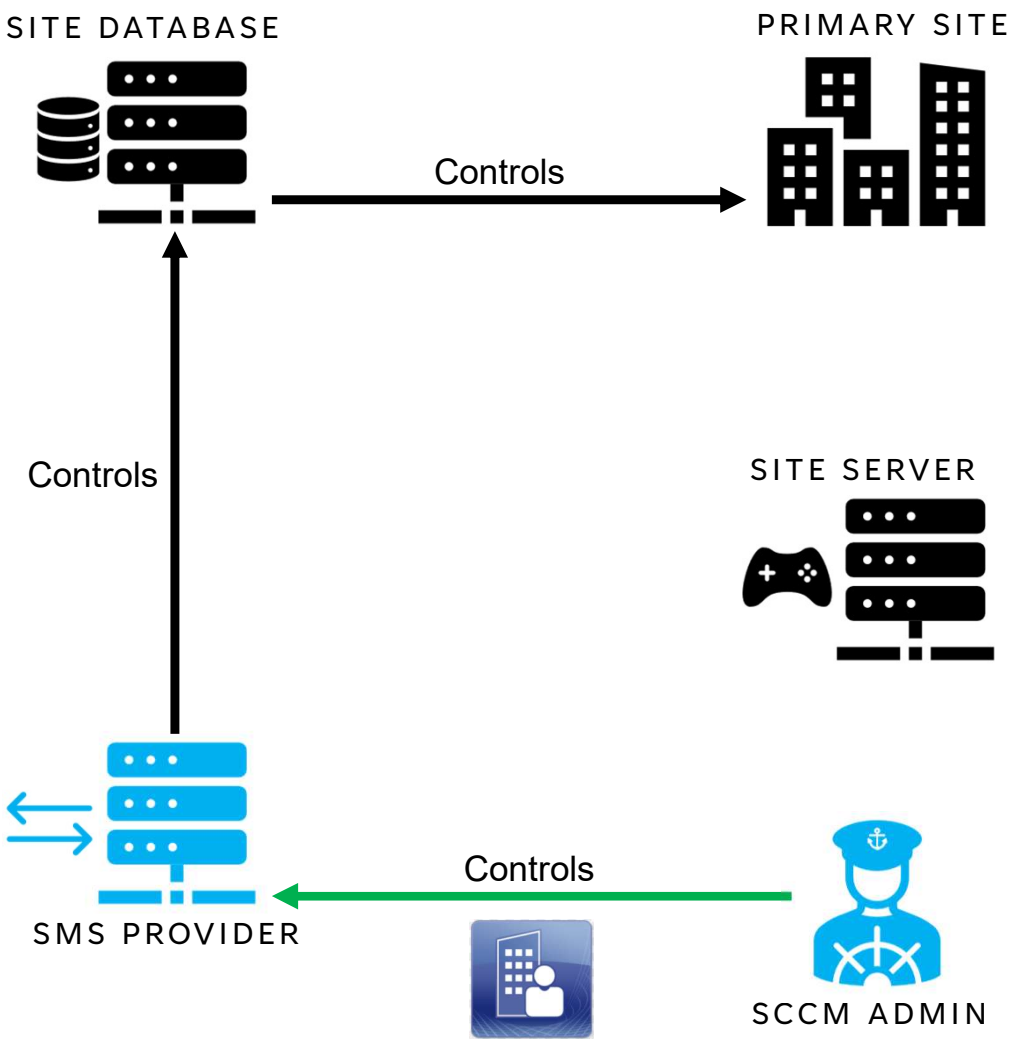
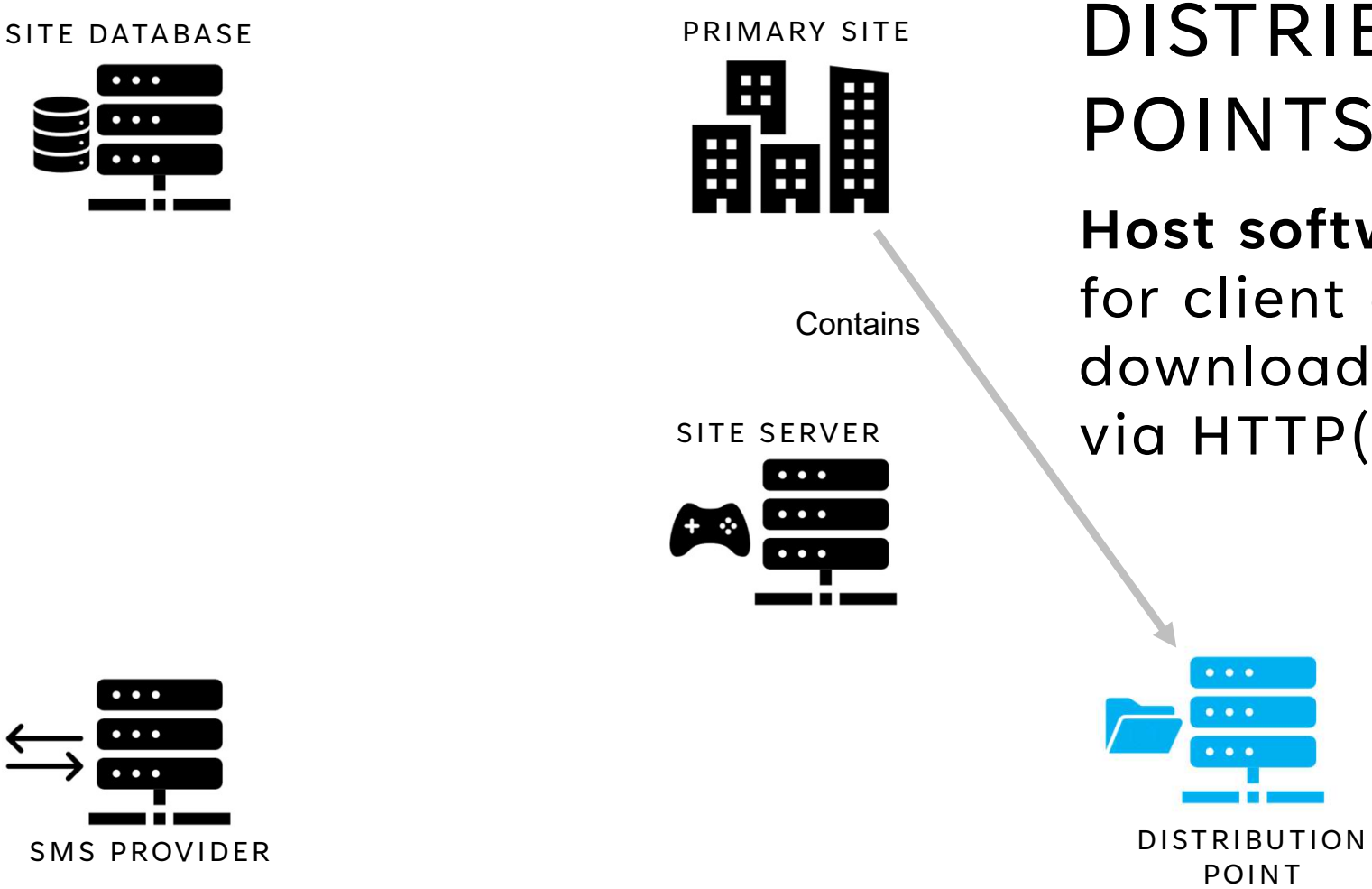


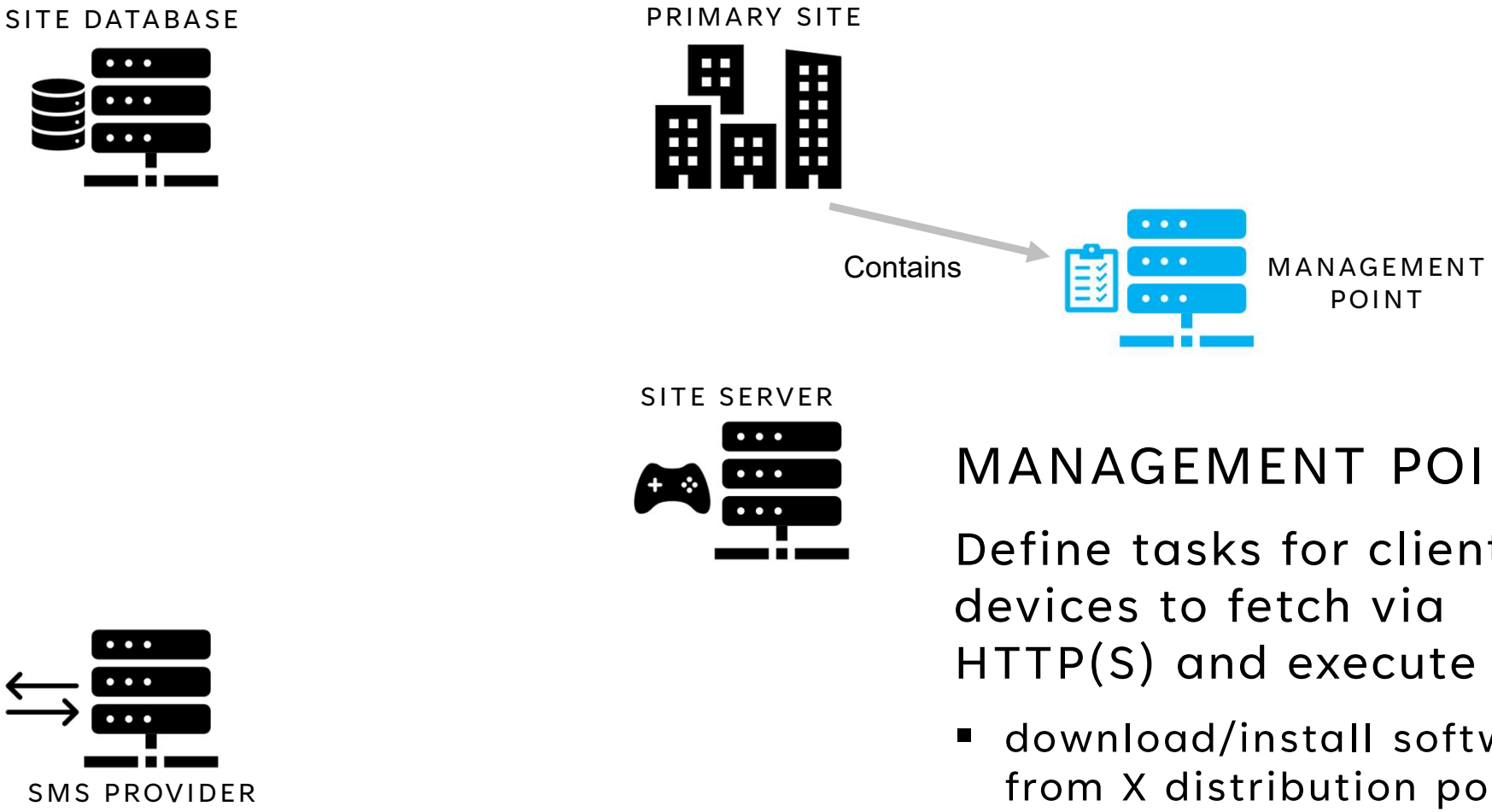
SCCM ADMINS

Use Configuration Manager **Console** software to control SCCM via SMS Provider APIs

DISTRIBUTION POINTS

Host software files for client devices to download and install via HTTP(S) and SMB





MANAGEMENT POINTS

Define tasks for client devices to fetch via HTTP(S) and execute

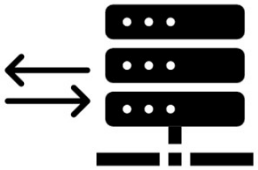
- download/install software from X distribution point

SITE DATABASE

PRIMARY SITE

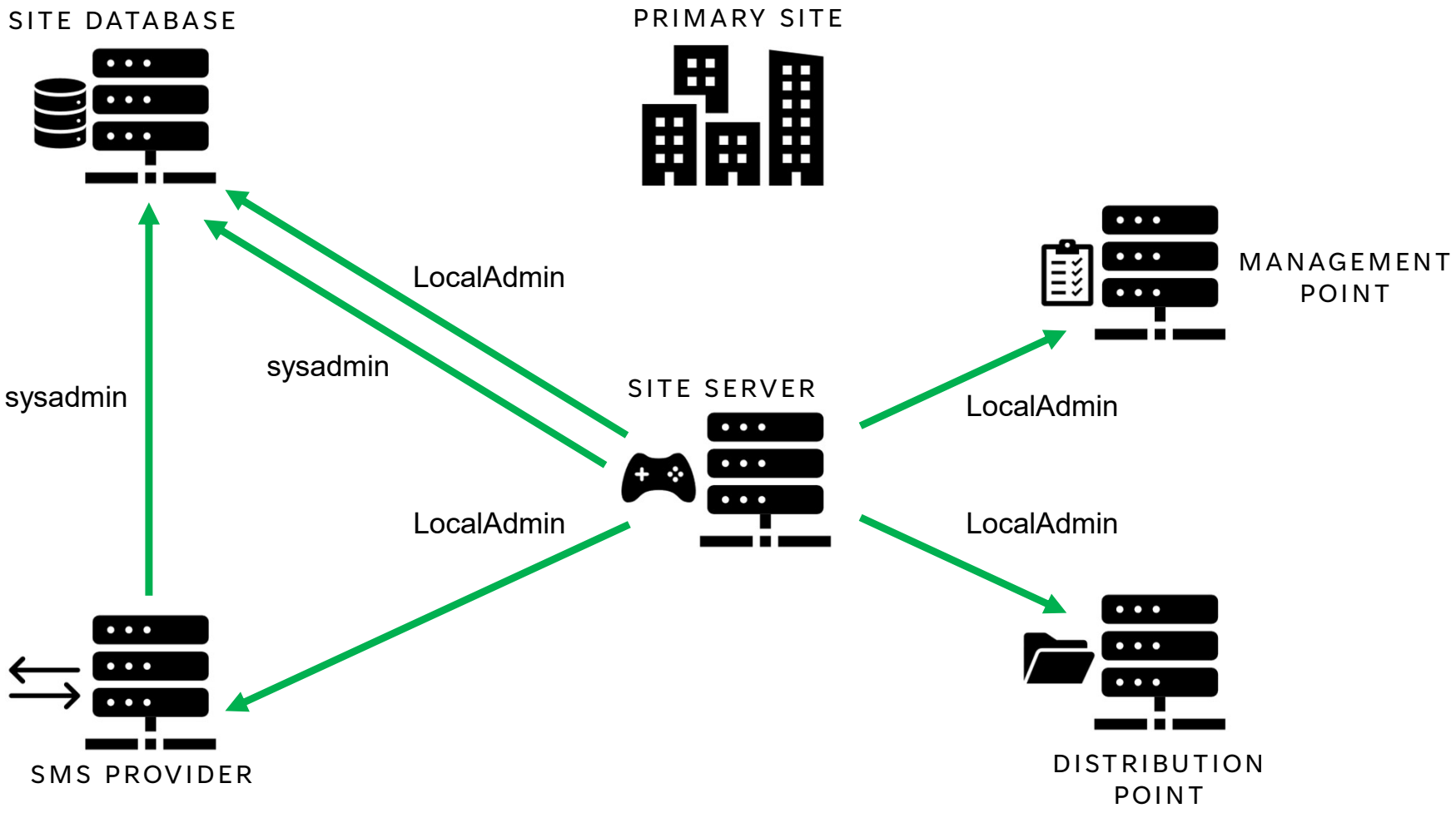
MANAGEMENT POINT

SITE SERVER



SMS PROVIDER

DISTRIBUTION POINT


SITE DATABASE
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LocalAdmin
sysadmin
sysadmin
SITE SERVER
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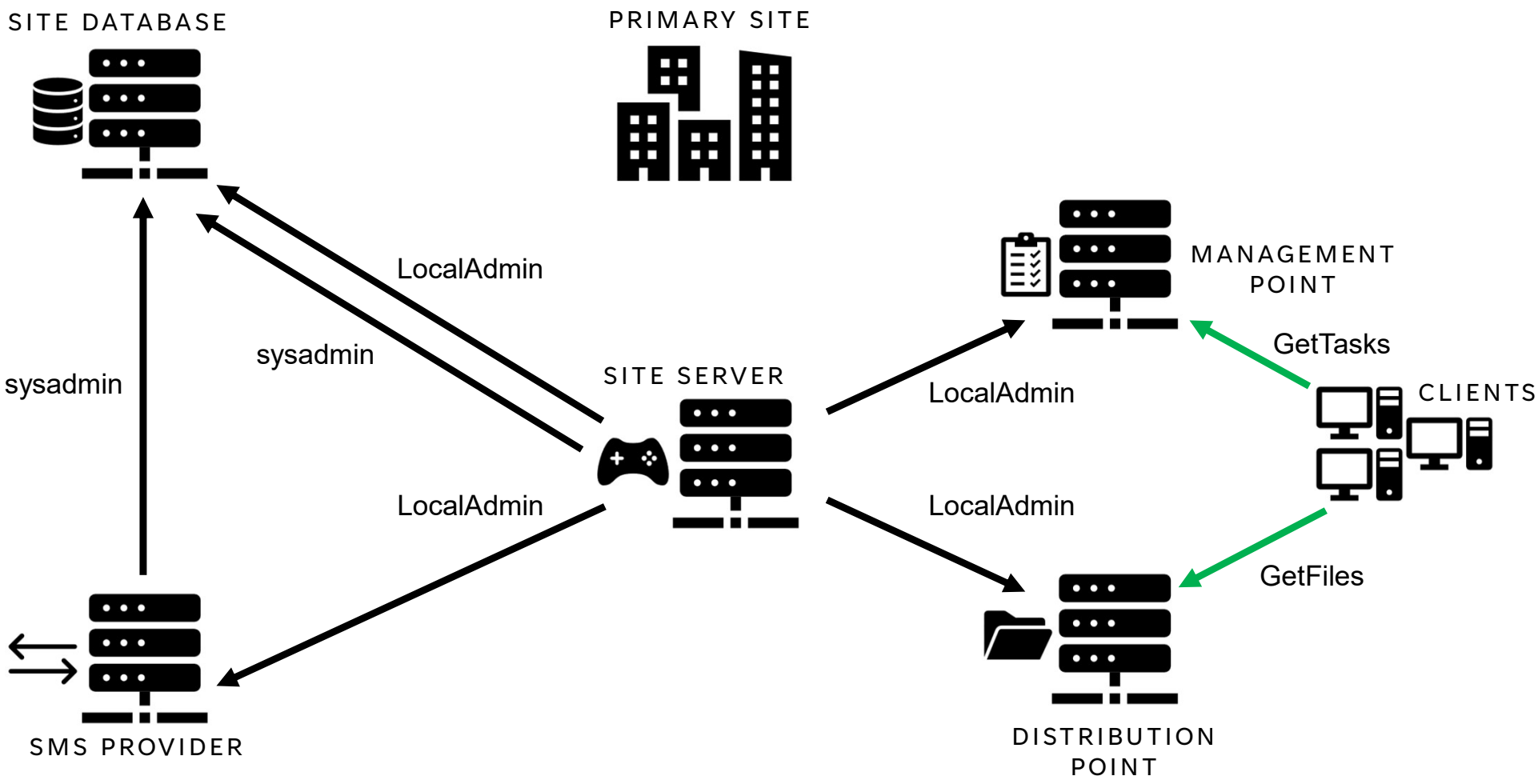
DISTRIBUTION POINT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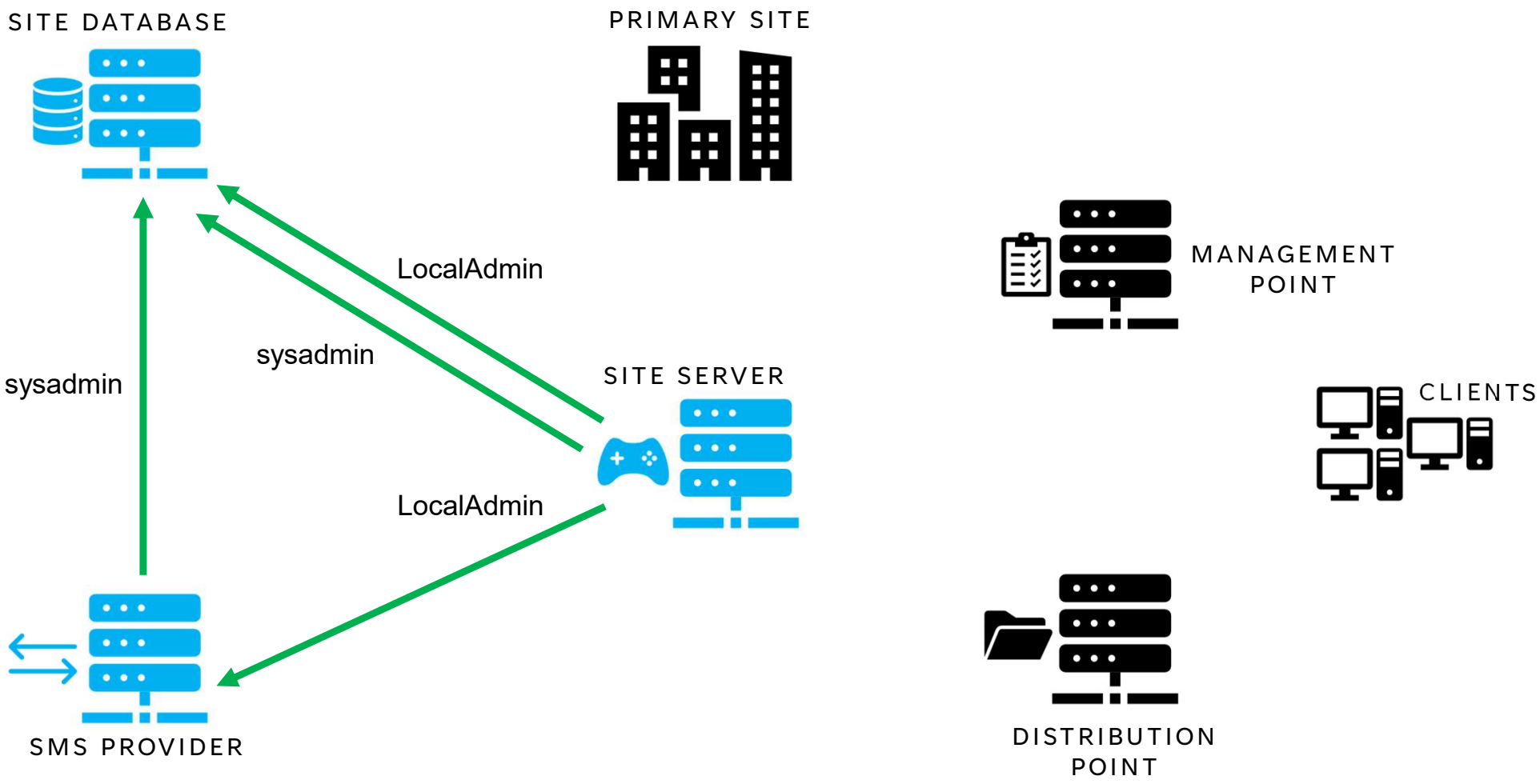
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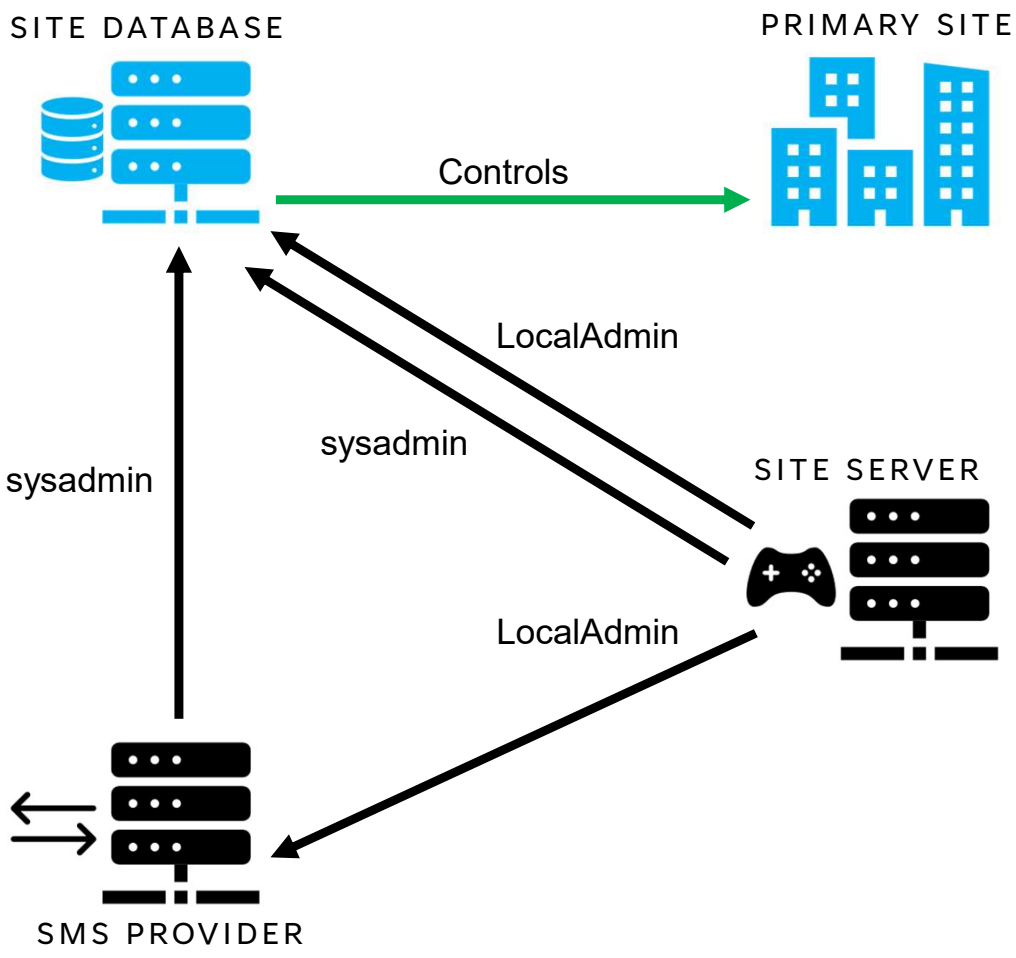
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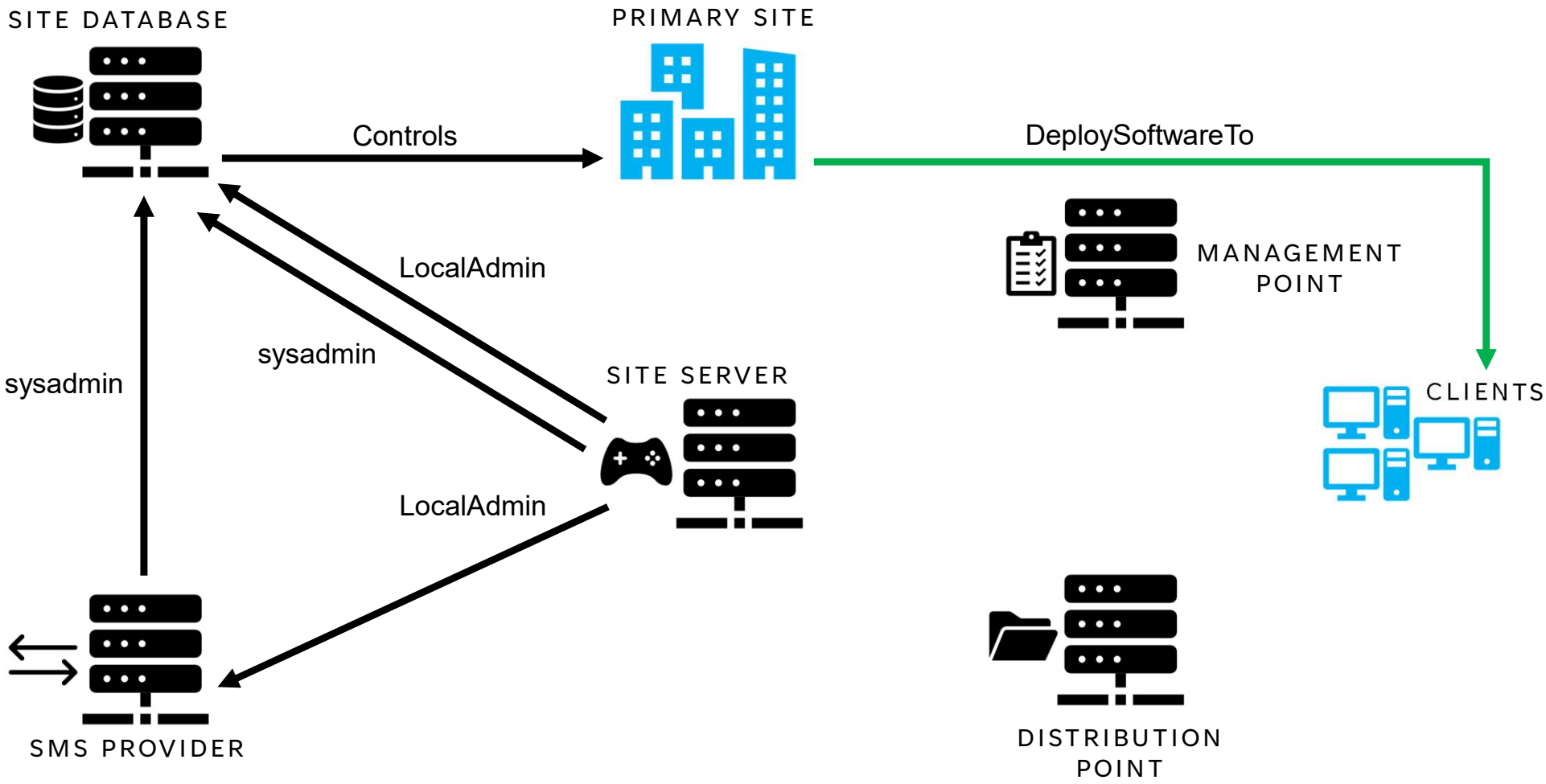
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LocalAdmin
SMS PROVIDER
DISTRIBUTION POINT

SCCM ATTACK PATHS

ALLOW CONTROL OF MANAGED DEVICES

33+ ATTACK TECHNIQUES

DISCOVERED SINCE 2022

SHOUTOUTS / PRIOR WORK

- Garrett Foster (@unsigned_sh0rt)
- Duane Michael (@subat0mik)
- Andy Robbins (@_wald0)
- Jonas Bülow Knudsen (@Jonas_B_K)
- Elad Shamir (@eladshamir)
- Zach Stein (@synzack21)
- Erik Hunstad (@badsectorlabs)
- Dylan Bradley (@slygoo)
- Nick Powers (@zynergy)
- Matt Creel (@Tw1sm)
- Lee Chagolla-Christensen (@tifkin_)
- Will Schroeder (@harmj0y)
- Adam Chester (@_xpn_)
- Christopher Panayi (@Raiona_ZA)
- Carsten Sandker (@0xcsandker)
- Josh Prager (@Praga_Prag)
- Mehdi Elyassa

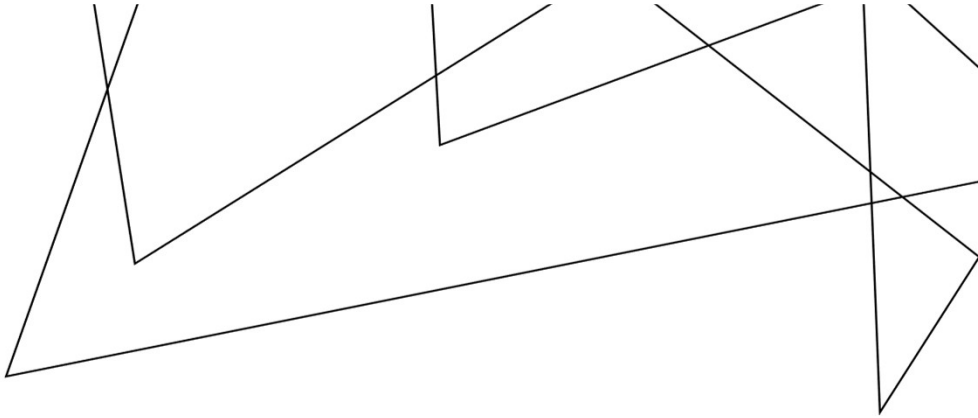
MISCONFIGURATION MANAGER

- Step-by-step offensive and defensive **write-ups** for attack techniques
- A **taxonomy** to simplify and demystify concepts
- Based on MITRE ATT&CK
- Co-authored with Duane Michael (@subat0mik) and Garrett Foster (@unsigned_sh0rt)

RECON

RECON

CRED



RECON

CRED

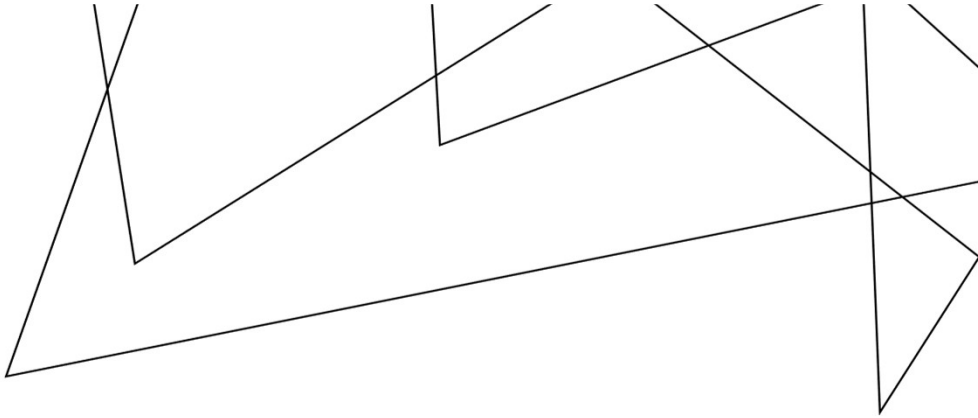
<u>COERCE</u> 😈

RECON

CRED

COERCE

<u>ELEVATE</u>



RECON

CRED

COERCE

ELEVATE

<u>TAKEOVER</u>

Grant a user the "**Full Administrator**" role by modifying SCCM via access to:

- site database
- SMS Provider APIs

RECON

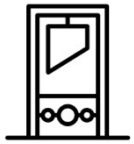
CRED

COERCE

ELEVATE

TAKEOVER

<u>EXEC</u>



ELEVATE

TAKEOVER

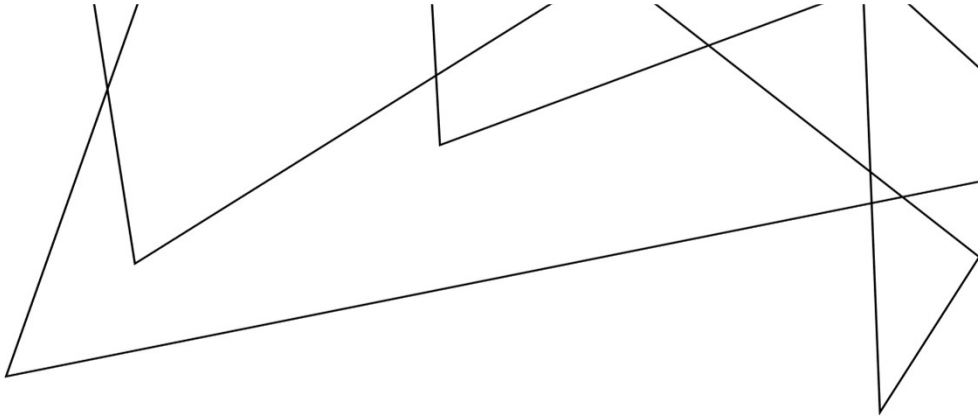
The site server's domain computer account MUST be:

- a **local Administrator** of every other SCCM server
- a **sysadmin** in the site database

NTLM

A protocol that Windows users and computers use to **authenticate over the network** without a password

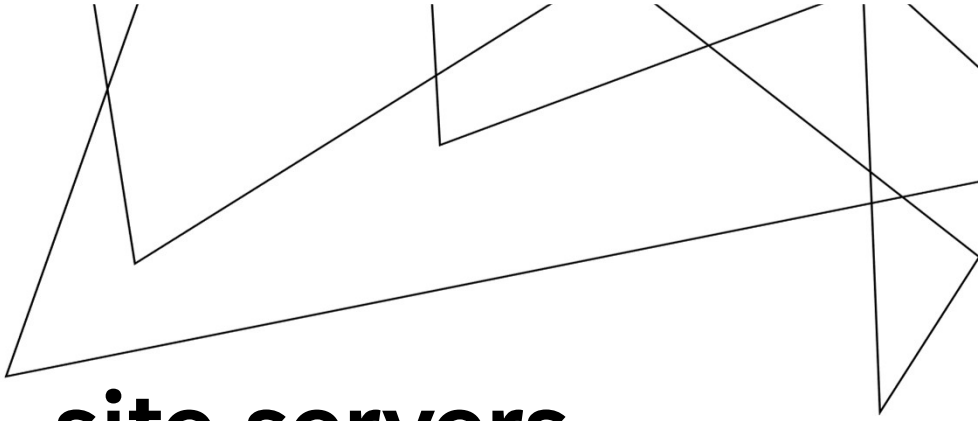
NTLM RELAY



Attackers can **coerce** computers into authenticating to their machine and **use** the creds on another machine where that computer has privileges

- e.g., Printerbug, PetitPotam

NTLM RELAY



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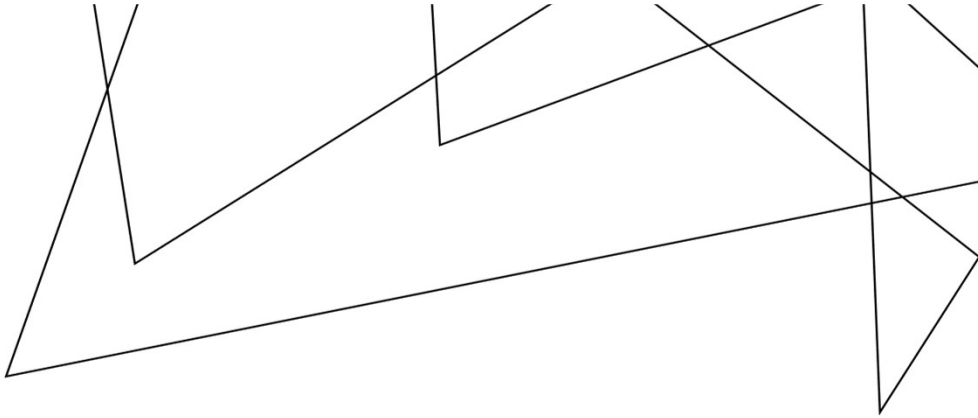
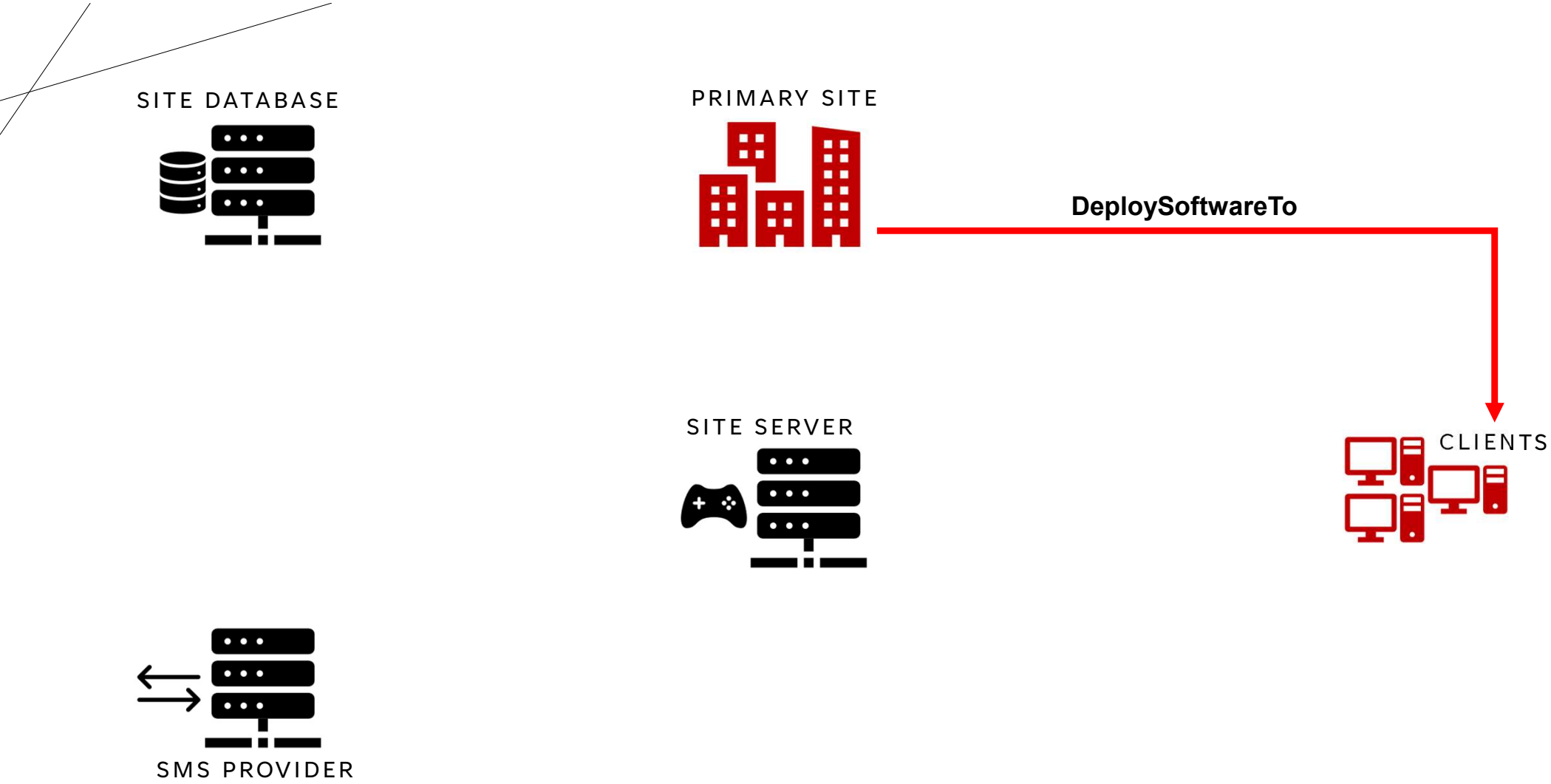


Susceptible when hosted **remotely** from the site server:

- site database
- SMS Provider APIs
- other site system roles

TAKEOVER

PRIMARY SITE
GrantAccess
DeploySoftwareTo
CLIENTS
ATTACKER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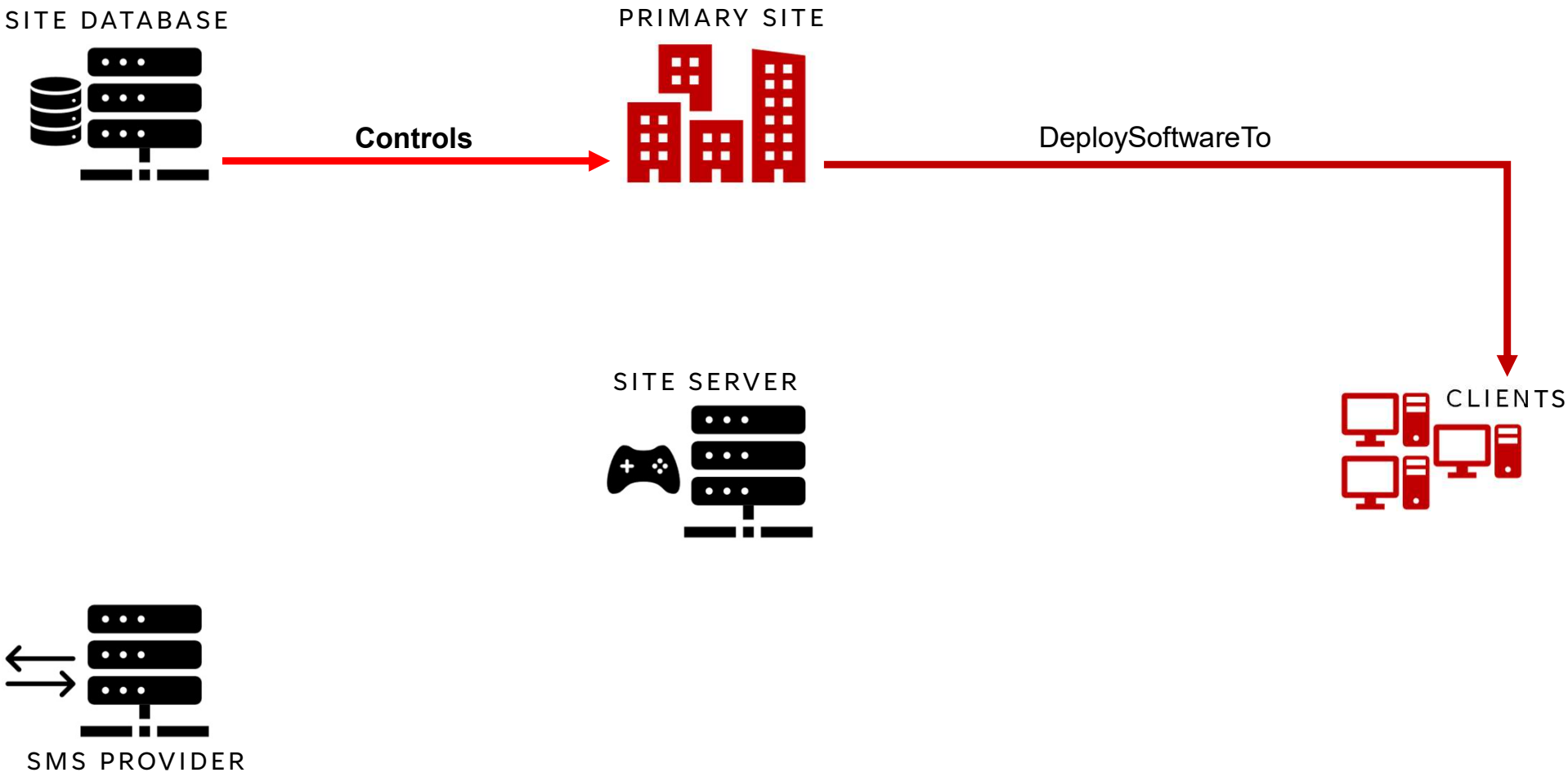
SITE DATABASE
Controls
PRIMARY SITE
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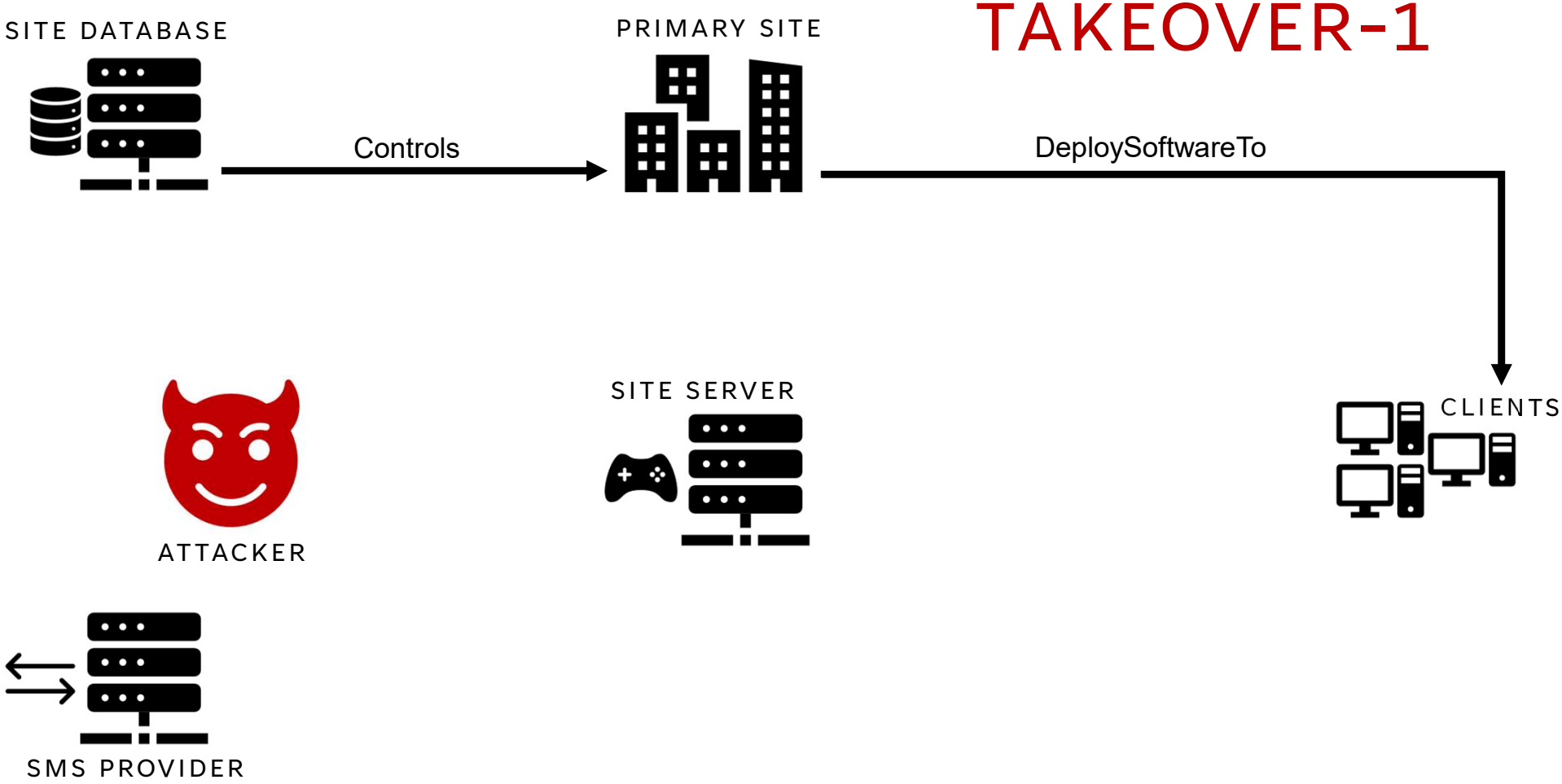
TAKEOVER-1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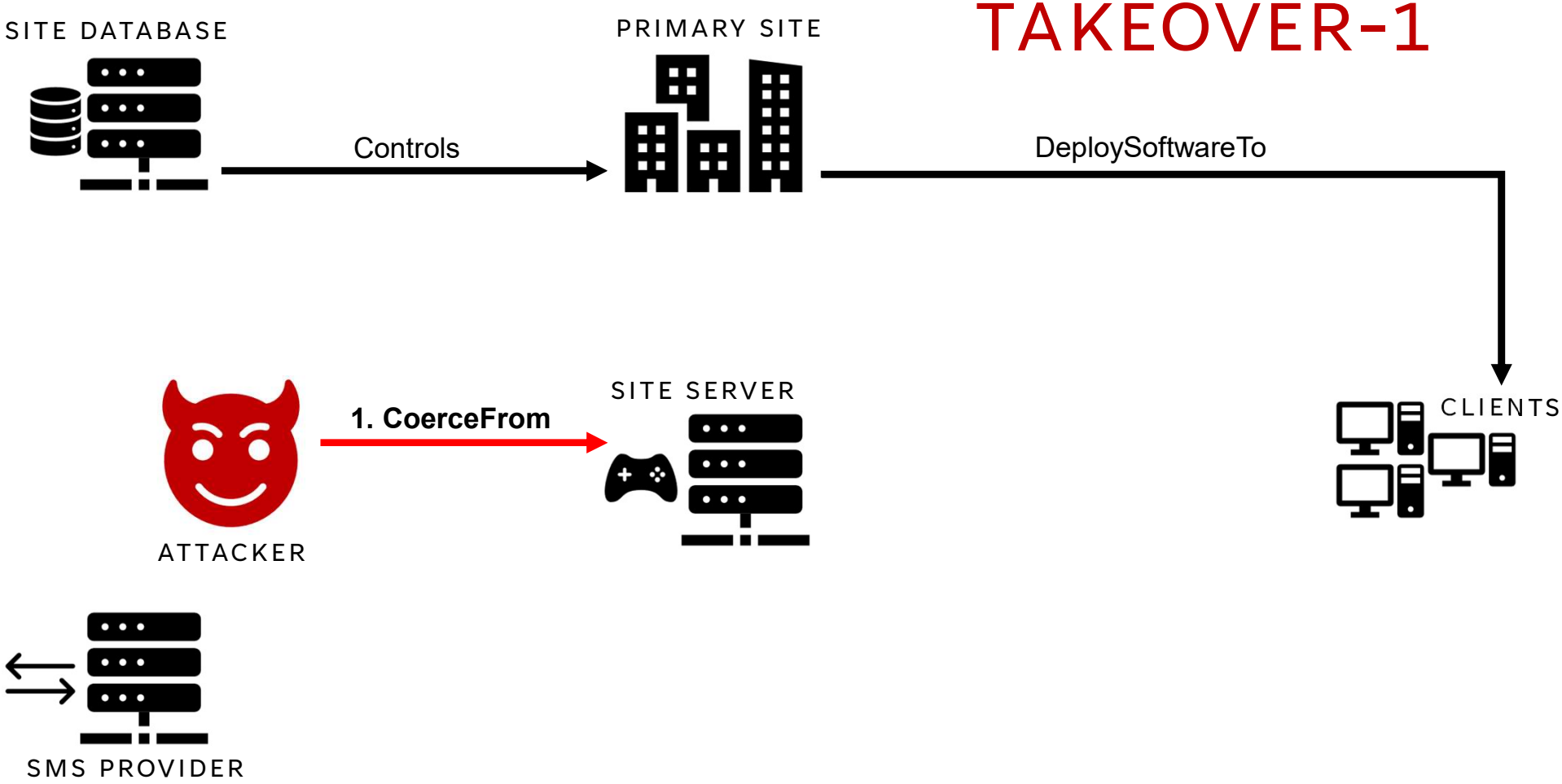
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1. CoerceFrom
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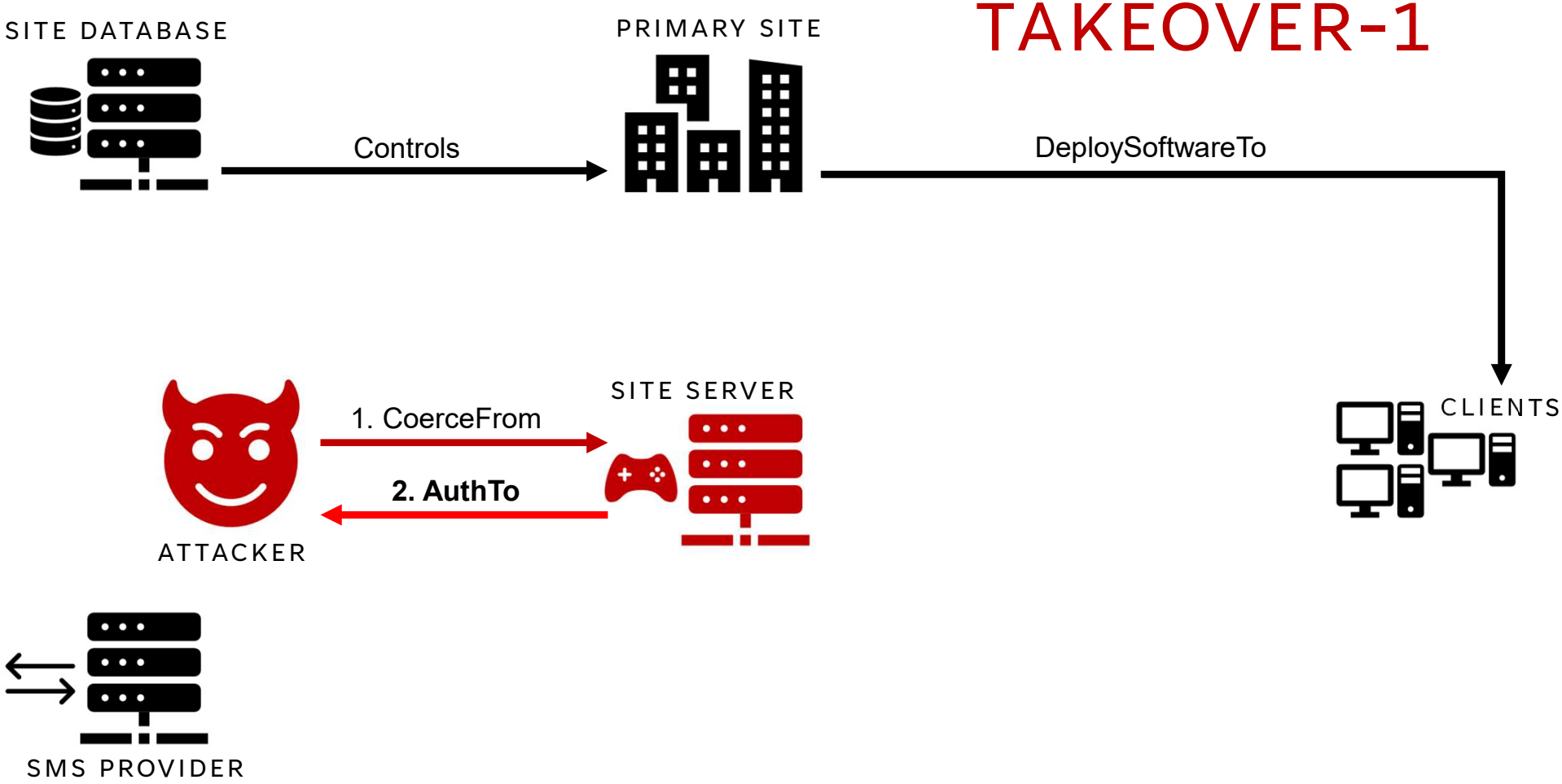
SITE DATABASE
Controls
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1. CoerceFrom
2. AuthTo
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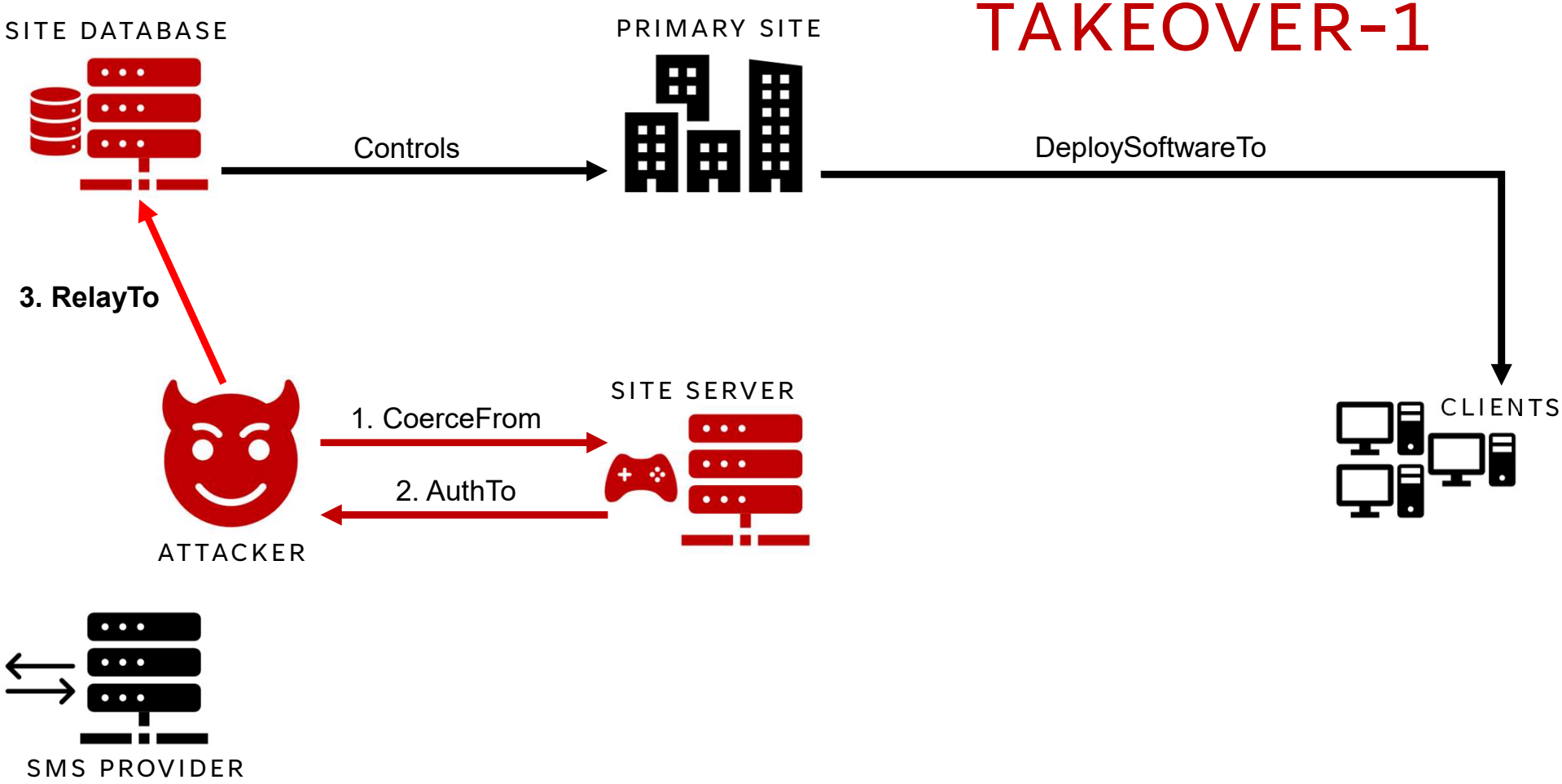
TAKEOVER-1


SITE DATABASE
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PRIMARY SITE
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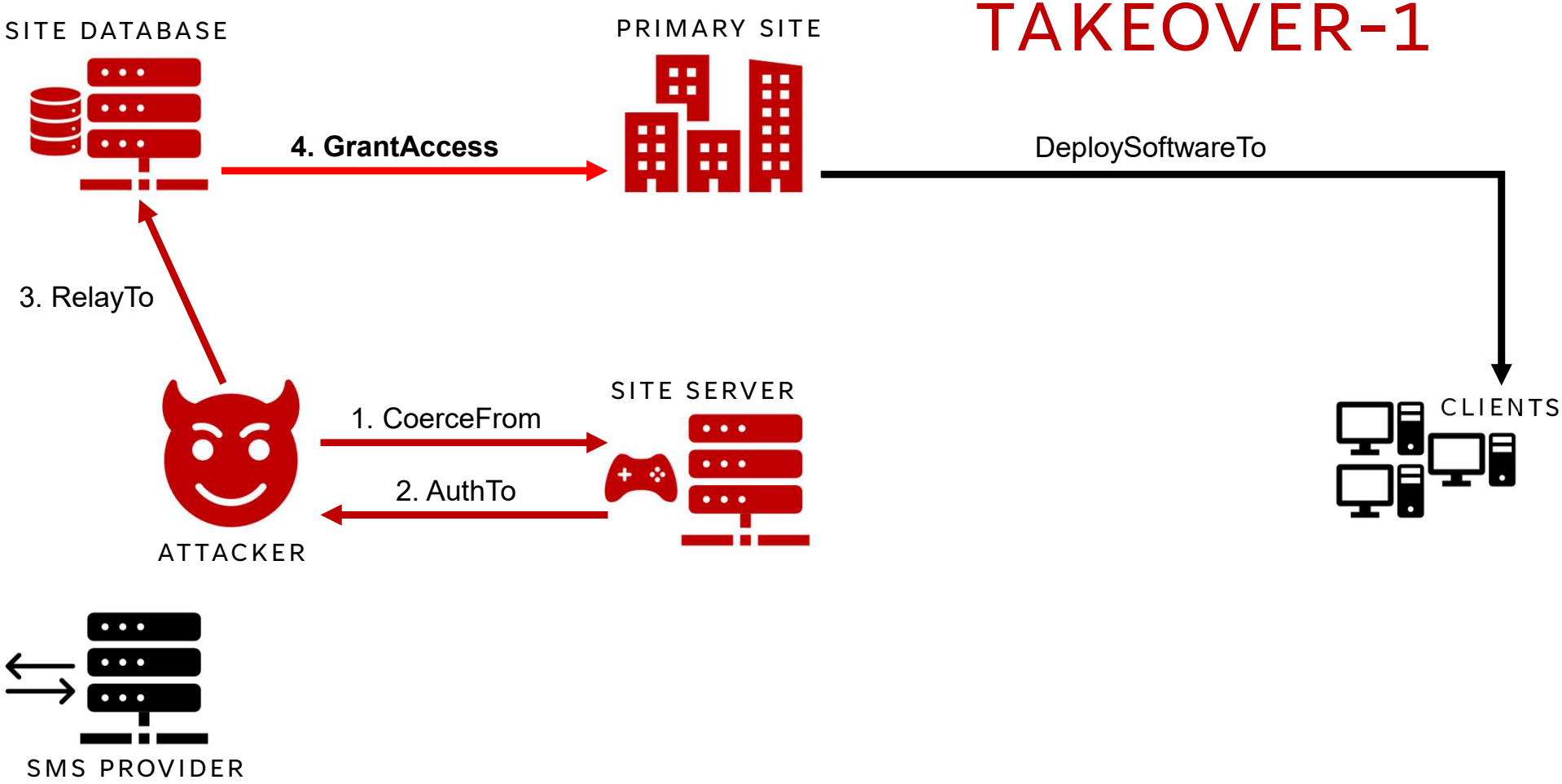
SITE DATABASE
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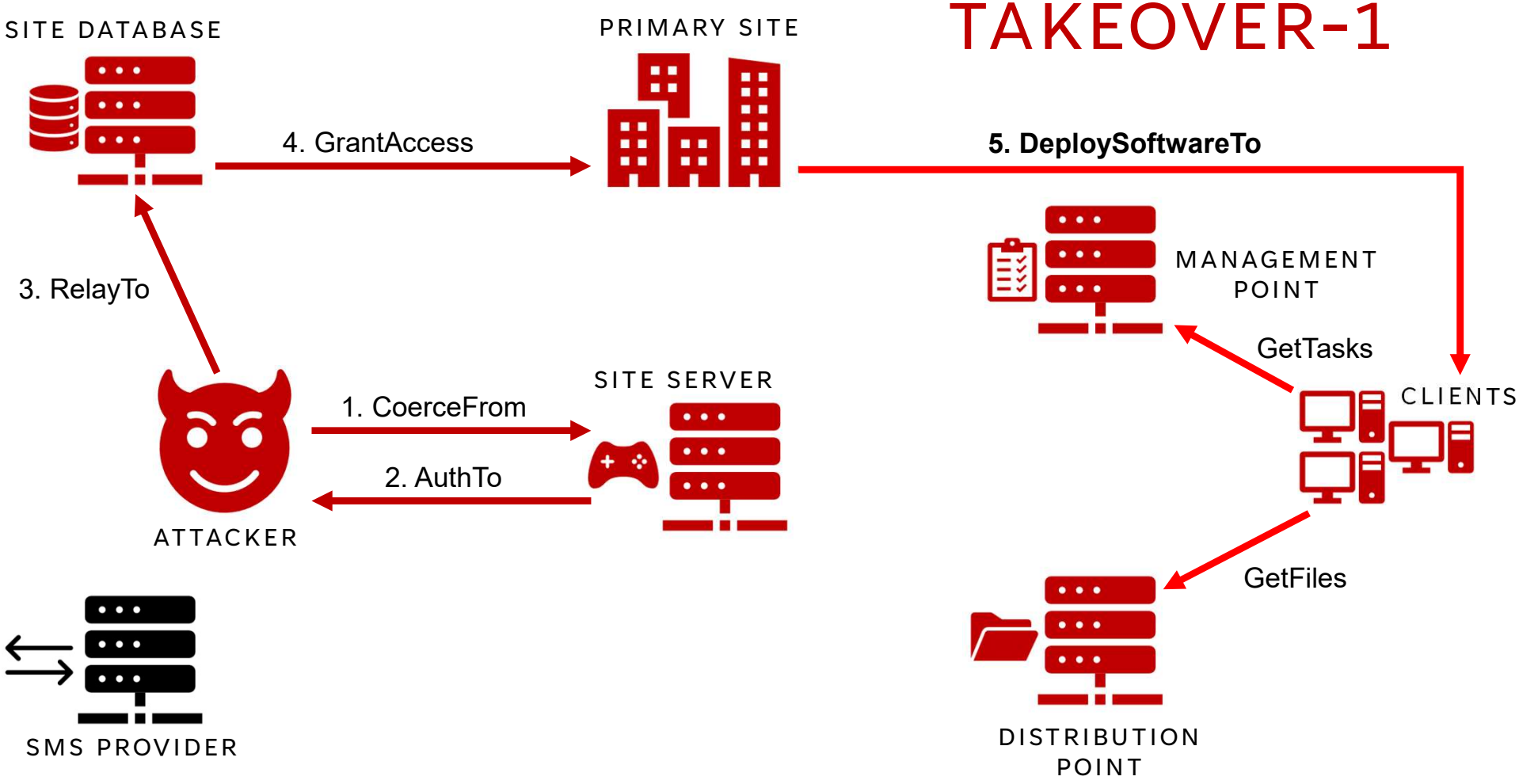
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GetTasks
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GetFiles
SMS PROVIDER
DISTRIBUTION POINT

TAKEOVER-1

PRIMARY SITE
GrantAccess
DeploySoftwareTo
CLIENTS
ATTACKER

SITE-SERVER - Remote Desktop

Microsoft Configuration Manager (Connected to PS1, Aperture Science - SITE-SERVER.APERTURE.LOCAL)

Home

Add User or Group | Saved Searches

Create | Search

One or more Azure AD app secrets used by Cloud Services have expired. Renew to avoid service disruptions. Renew expired secret key 1/1

\ ▸ Administration ▸ Overview ▸ Security ▸ Administrative Users

Administration

- Overview
  - Updates and Servicing
  - Hierarchy Configuration
  - Cloud Services
  - Site Configuration
  - Client Settings
  - Security

Assets and Compliance

Software Library

Monitoring

Administration

Community

Administrative Users 1 items

Search current node | Search | Add Criteria

Icon	Account Name	Account Display Name	Security Roles
	APERTURE\labadmin		"Full Administrator"

Ready

Type here to search

9:30 PM
1/29/2026

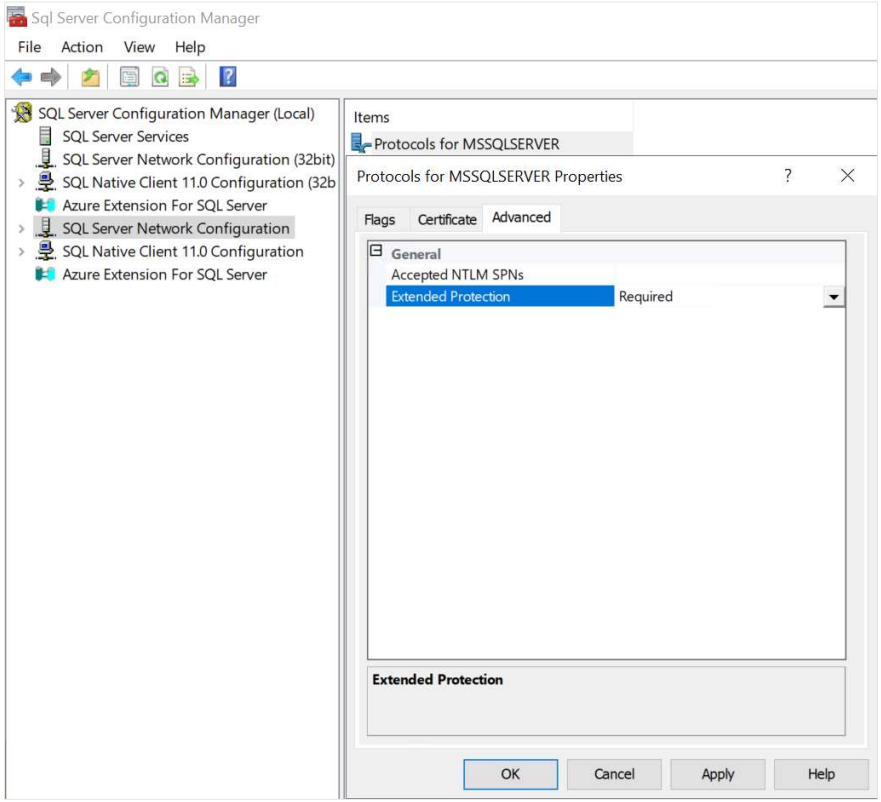
TAKEOVER-1 REMEDIATION

Require Extended Protection On Remote Site Databases

On **remote** site database servers:

1. Open Sql Server Configuration Manager
2. Click “Sql Server Network Configuration”
3. Right click “Protocols for MSSQLSERVER”, then click “Properties”
5. Navigate to the “Advanced” tab
6. Set “Extended Protection” to “Required”
7. Click “Apply”, then “OK”
8. Restart the “SQL Server (MSSQLSERVER)” service

Setting to “Allowed” will not prevent NTLM relay attacks if the coerced client doesn’t support channel binding



TAKEOVER-5

PRIMARY SITE
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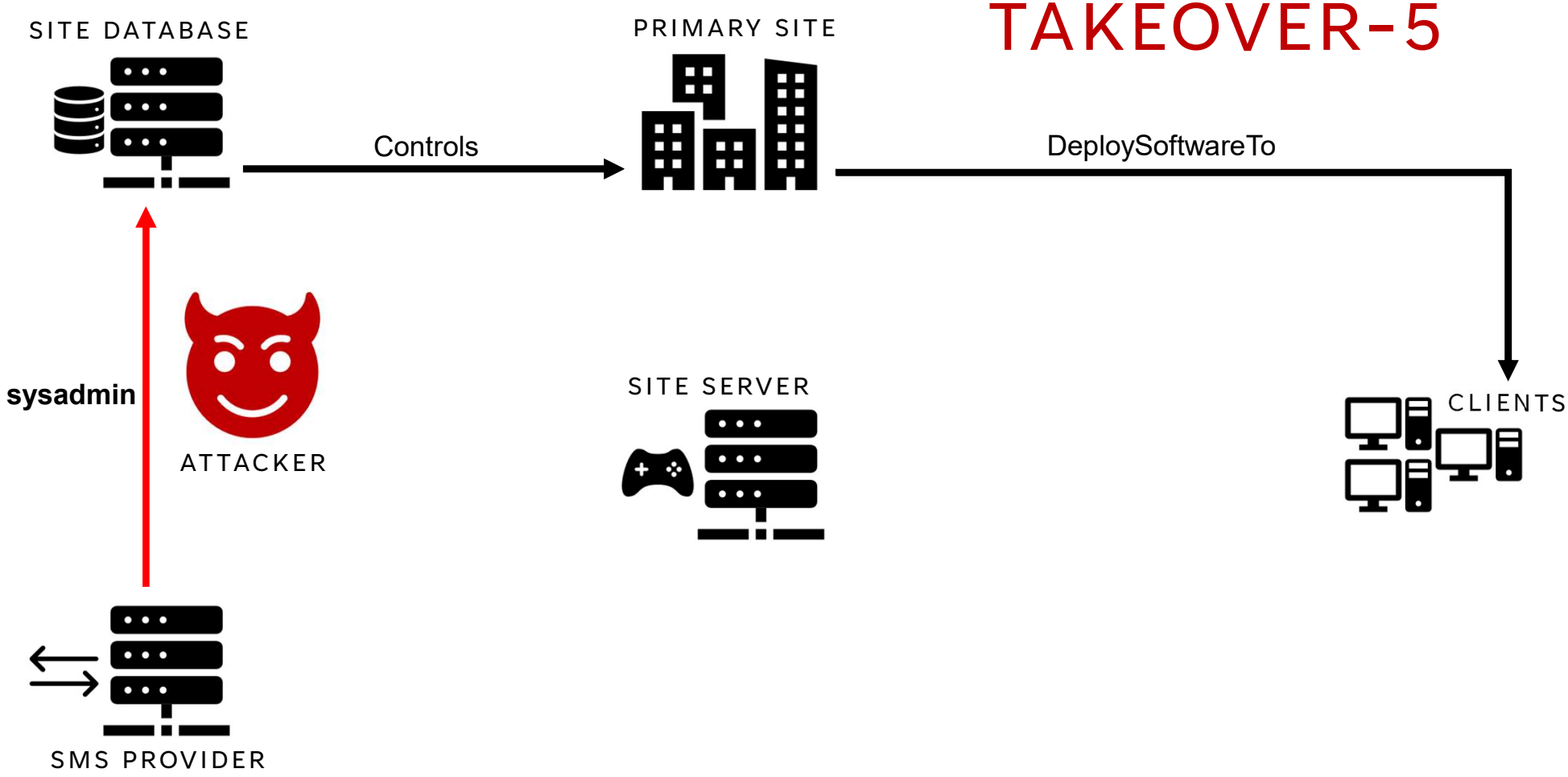
ATTACKER

TAKEOVER-5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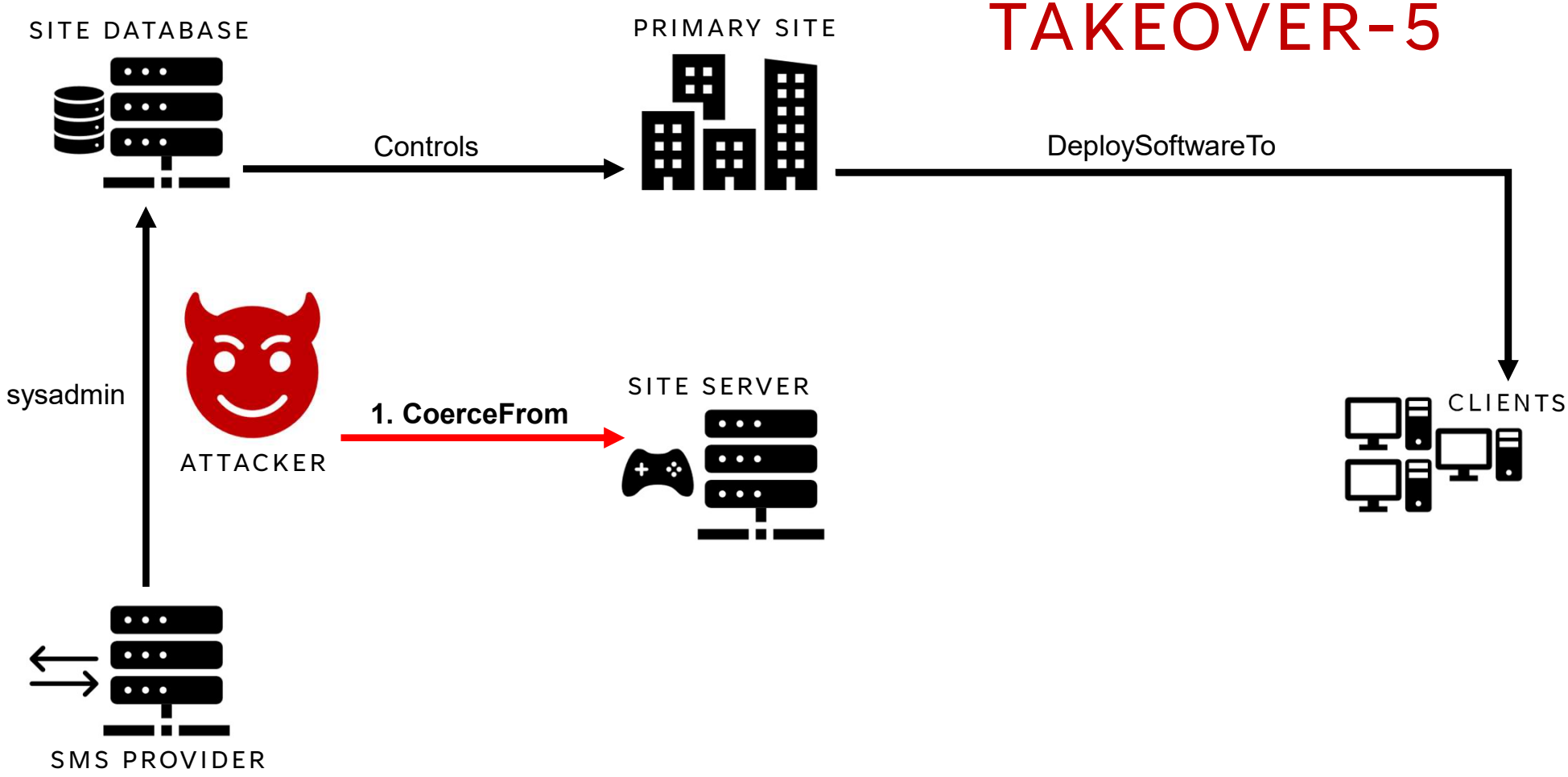
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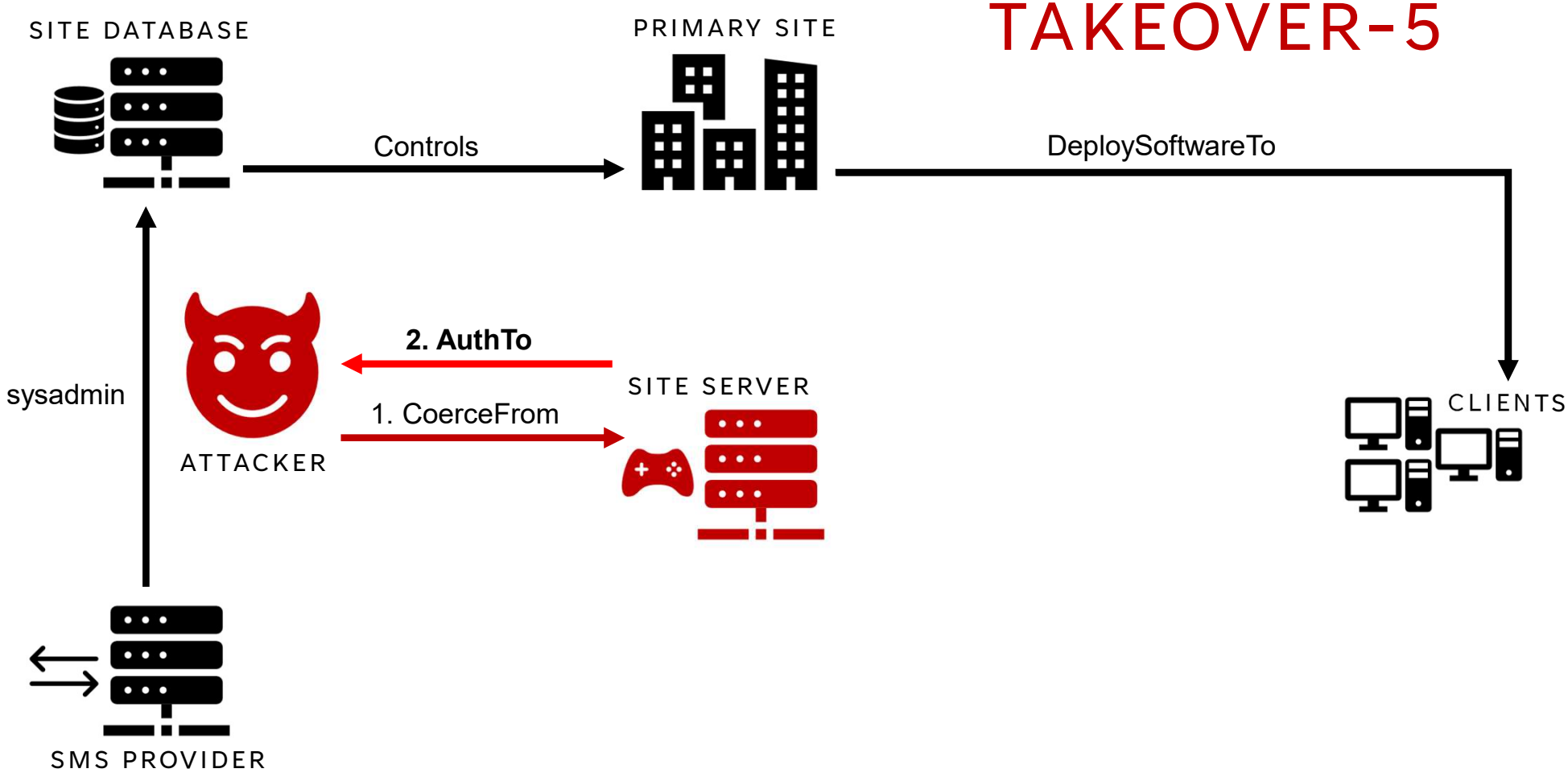
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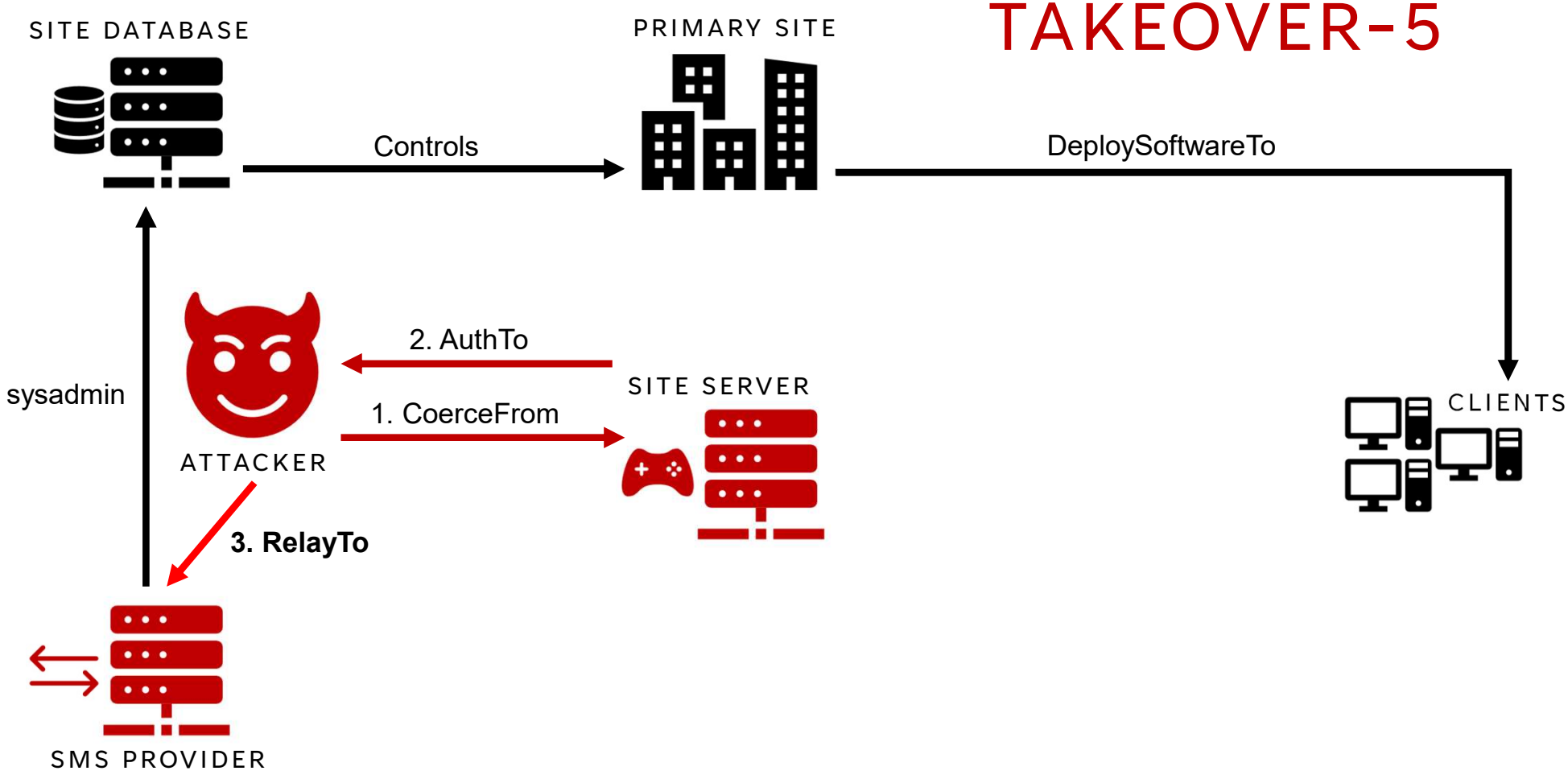
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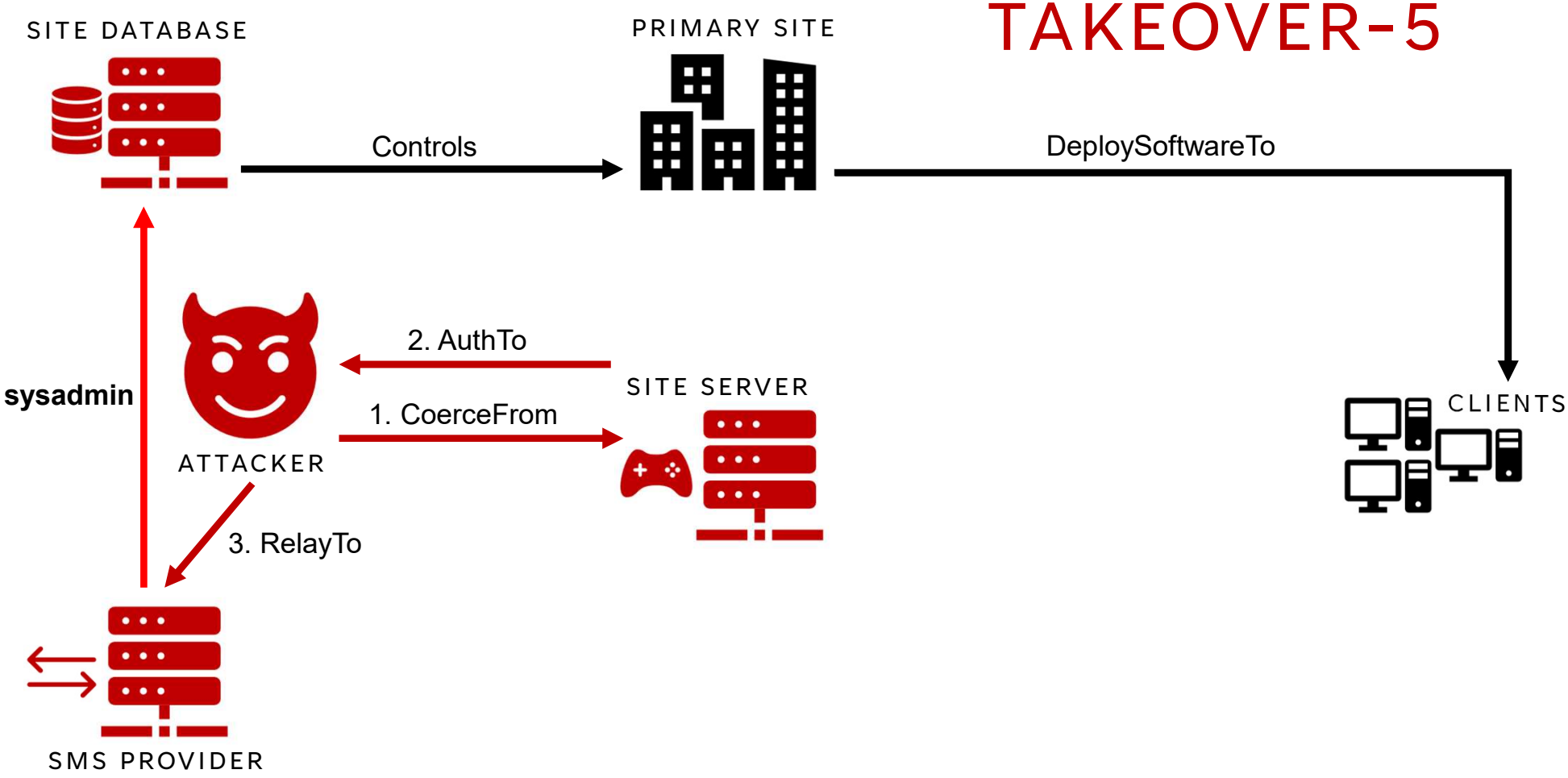
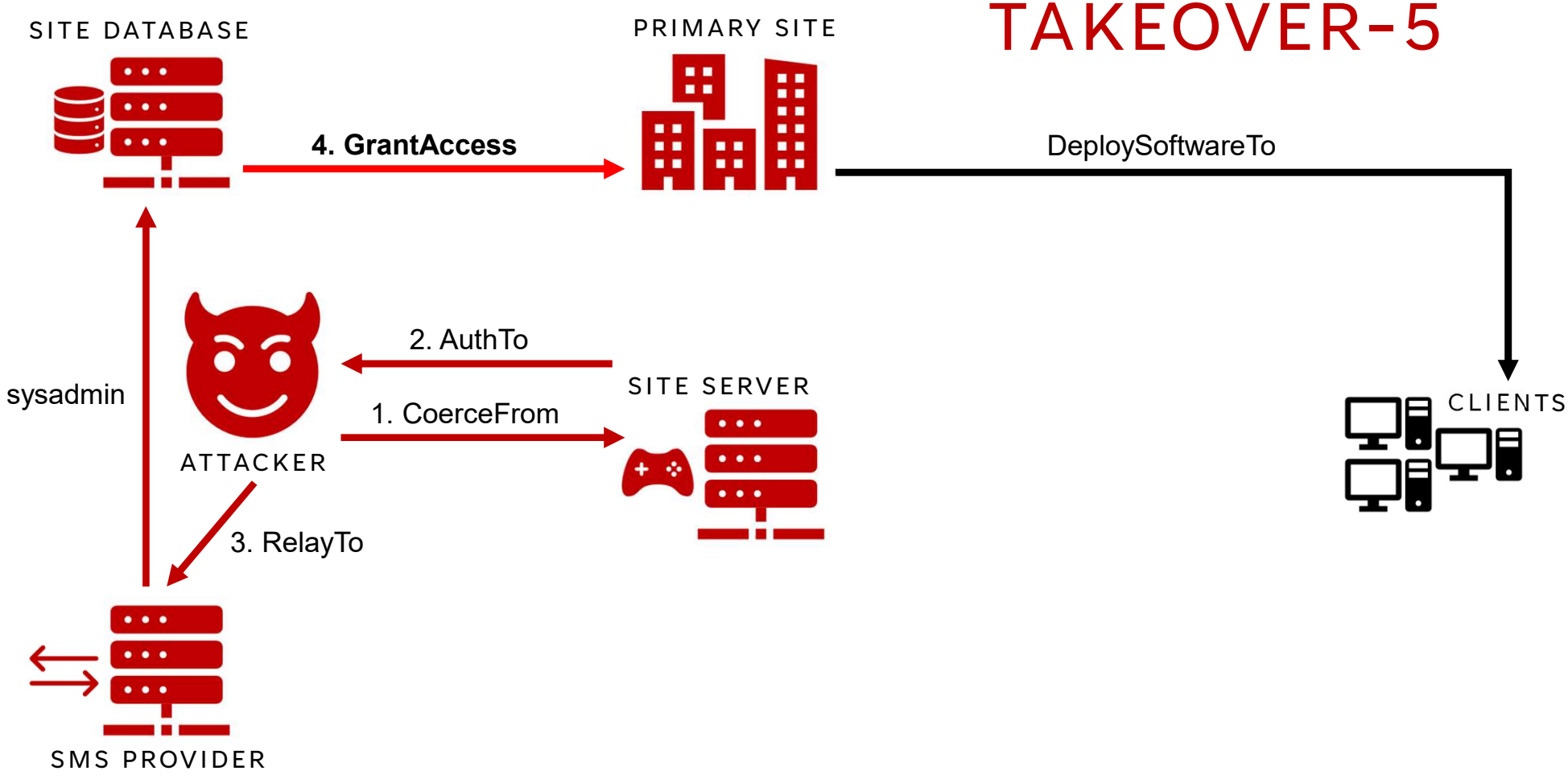
3. RelayTo
SMS PROVIDER

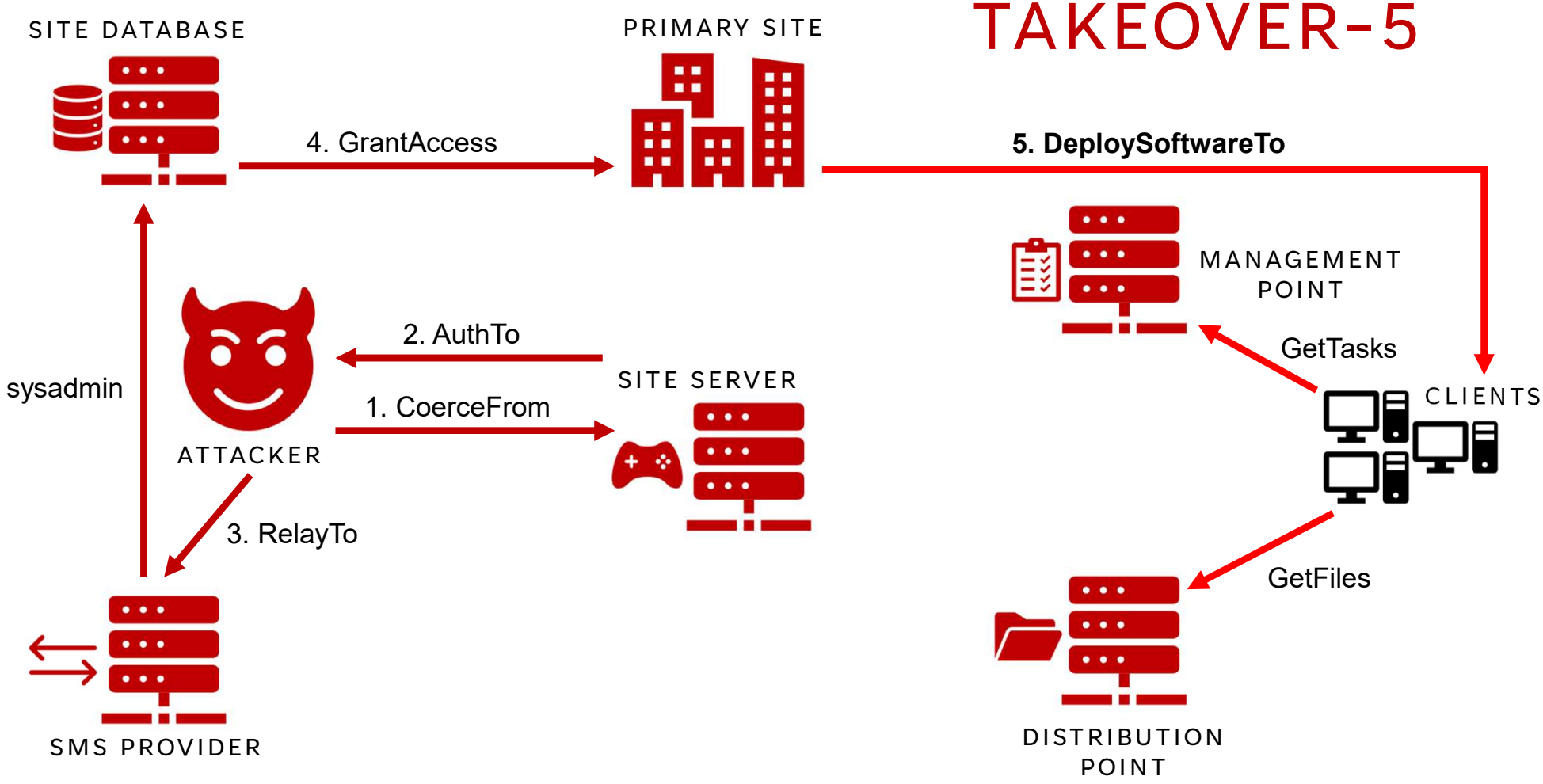
TAKEOVER-5


SITE DATABASE
PRIMARY SITE
Controls
DeploySoftwareTo
sysadmin
2. AuthTo
1. CoerceFrom
SITE SERVER
ATTACKER
CLIENTS
3. RelayTo
SMS PROVIDER

TAKEOVER-5


SITE DATABASE
PRIMARY SITE
4. GrantAccess
DeploySoftwareTo
2. AuthTo
SITE SERVER
sysadmin
1. CoerceFrom
CLIENTS
ATTACKER
3. RelayTo
SMS PROVIDER

TAKEOVER-5


SITE DATABASE
PRIMARY SITE
4. GrantAccess
5. DeploySoftwareTo
MANAGEMENT POINT
GetTasks
CLIENTS
2. AuthTo
SITE SERVER
sysadmin
1. CoerceFrom
ATTACKER
3. RelayTo
GetFiles
SMS PROVIDER
DISTRIBUTION POINT

TAKEOVER-5

PRIMARY SITE
GrantAccess
DeploySoftwareTo
CLIENTS
ATTACKER

SITE-SERVER - Remote Desktop

Microsoft Configuration Manager (Connected to PS1, Aperture Science - SITE-SERVER.APERTURE.LOCAL)

Home

Add User or Group | Saved Searches

Create | Search

One or more Azure AD app secrets used by Cloud Services have expired. Renew to avoid service disruptions. Renew expired secret key 1/1

\ ▸ Administration ▸ Overview ▸ Security ▸ Administrative Users

Administration

- Overview
  - Updates and Servicing
  - Hierarchy Configuration
  - Cloud Services
  - Site Configuration
  - Client Settings
  - Security
- Assets and Compliance
- Software Library
- Monitoring
- Administration
- Community

Administrative Users 0 items

Search current node | Search | Add Criteria

Icon	Account Name	Account Display Name	Security Roles
	APERTURE\labadmin		"Full Administrator"

APERTURE\labadmin

Ready

Type here to search

7:47 PM
1/29/2026

TAKEOVER-5 REMEDIATION

- Only host the SMS Provider role on the site server

OR

- **Upgrade** to Configuration Manager **v2509**, which denies NTLM authentication to the SMS Provider AdminService by default

OPENGRAPH

https://bloodhound.specterops.io/opengraph/library

BLOODHOUND

Attack path management for:

- Active Directory
- Azure AD / Entra ID
- All the things!

Free and open-source!

https://bloodhound.specterops.io/

OpenGraph Library

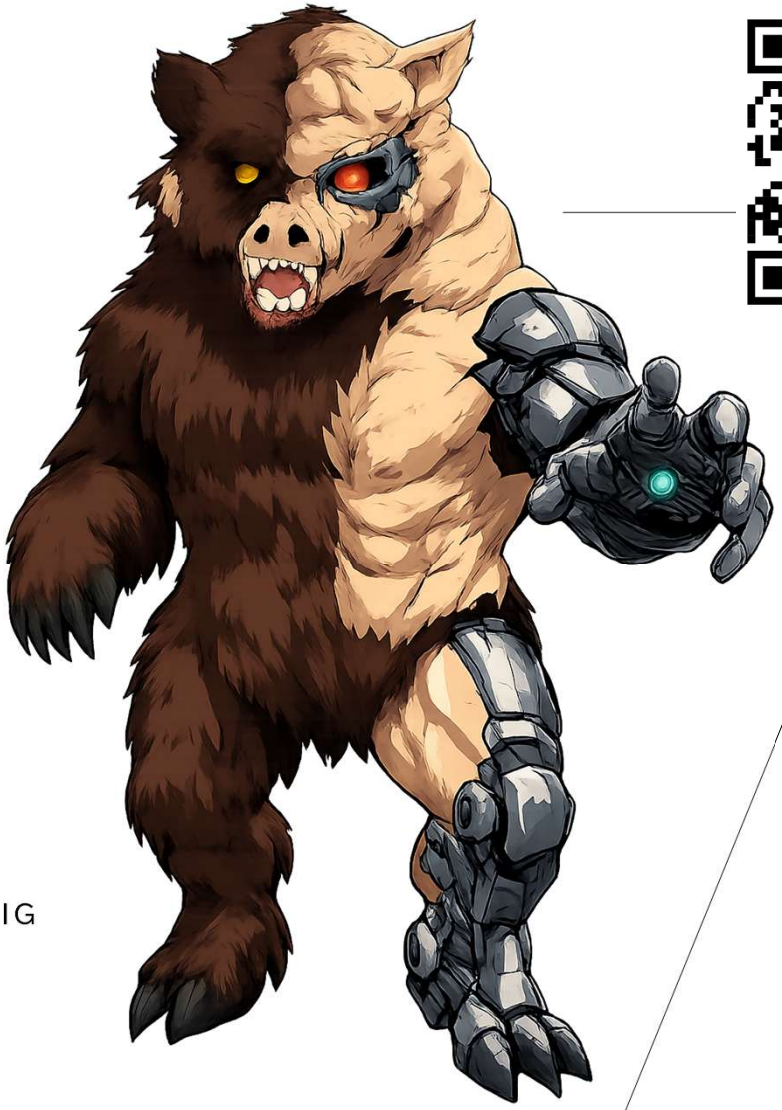
- 1Password
  - 1PassHound
- Ansible
  - AnsibleHound
- Active Directory (AD)
  - ADAttributeHound

Unfortunately...

The name **SCCMHOUND** was already taken

ConfigManBearPig

- A PowerShell (Python coming soon) OpenGraph collector for SCCM
- 5 new nodes
- 20 new edges
- 13 existing edges
- Does NOT require privileged access, just a domain user



HTTPS://GITHUB.COM/MAYYHEM/CONFIGMANBEARPIG

RECON: 6/7

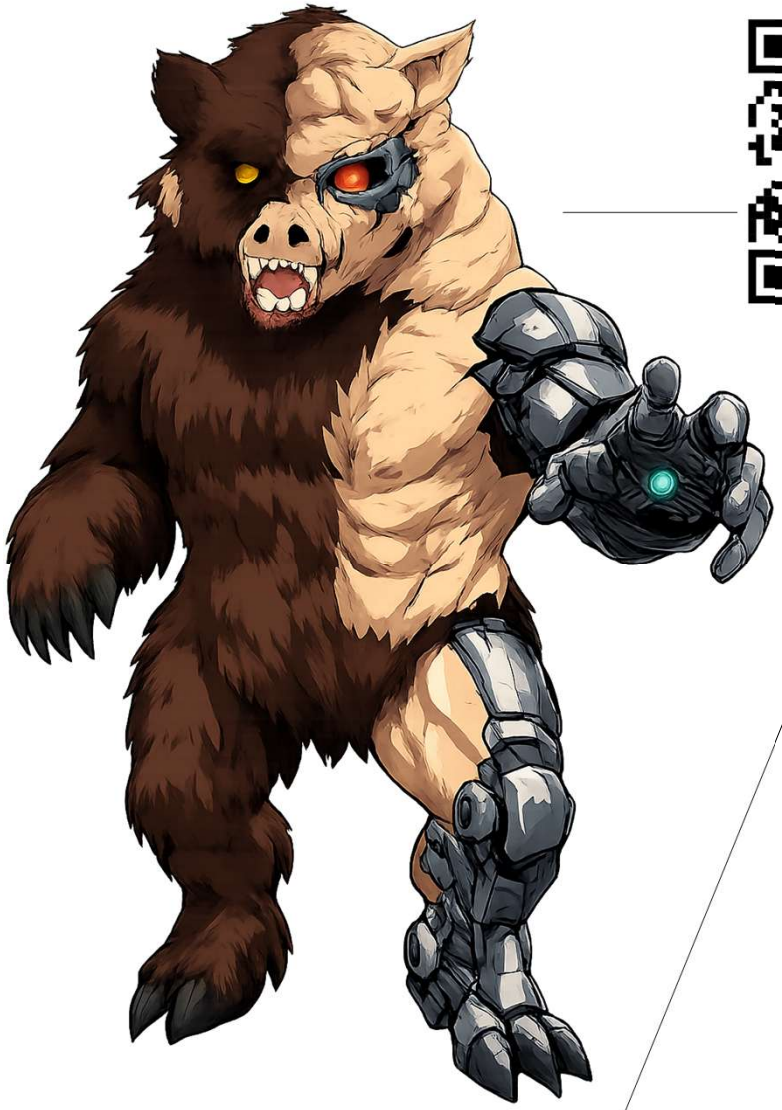
CRED: 3/8

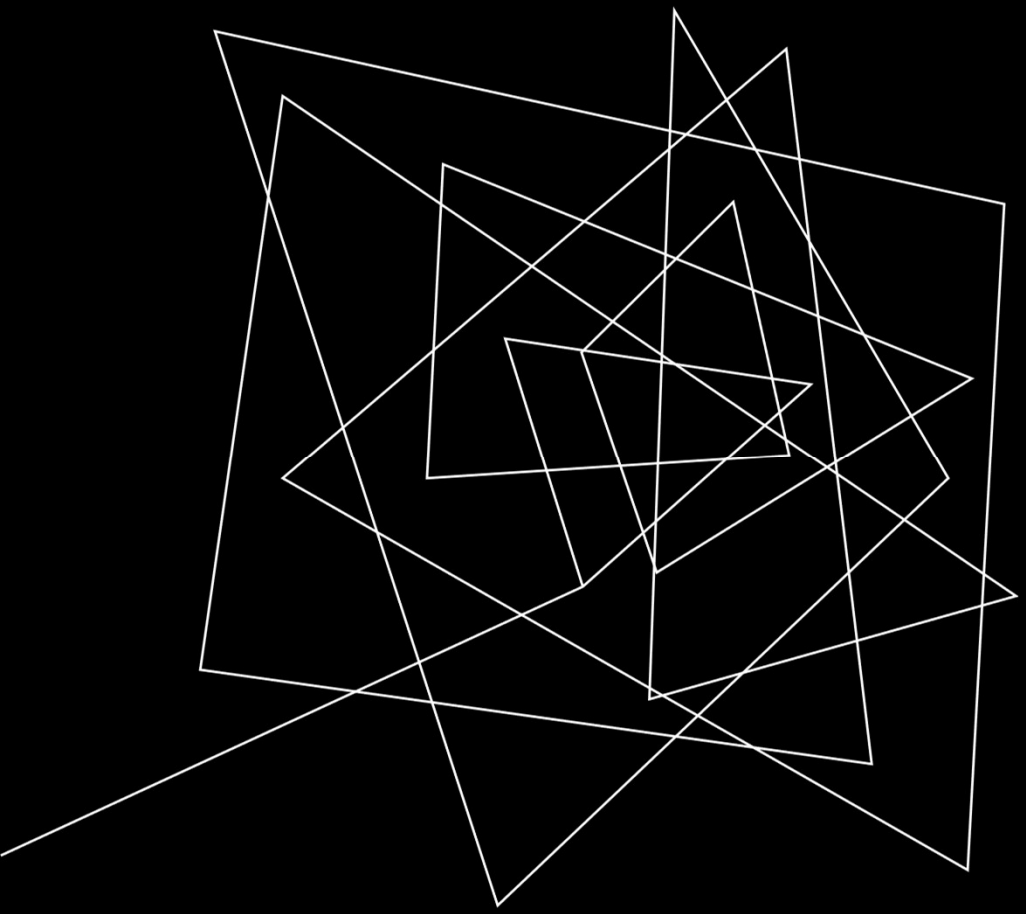
COERCE: 0/2

ELEVATE: 1/5

TAKEOVER: 9/9

EXEC: 2/2

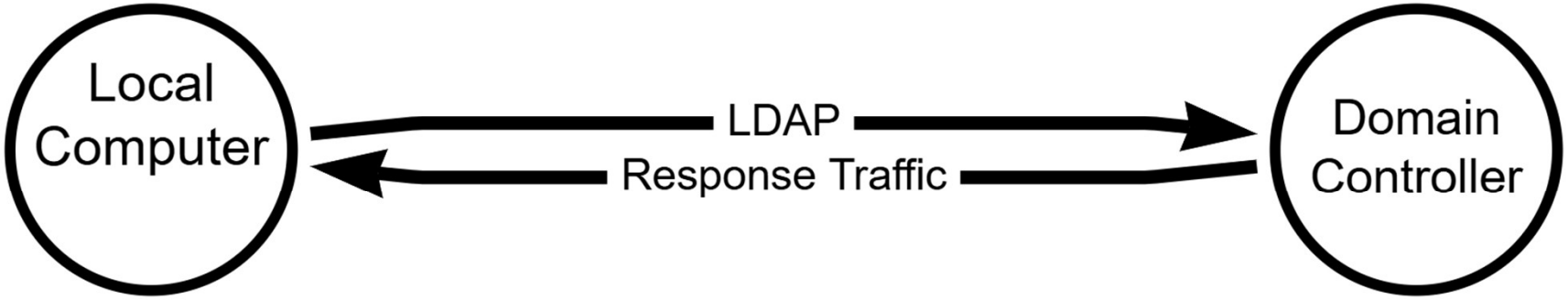




COLLECTION

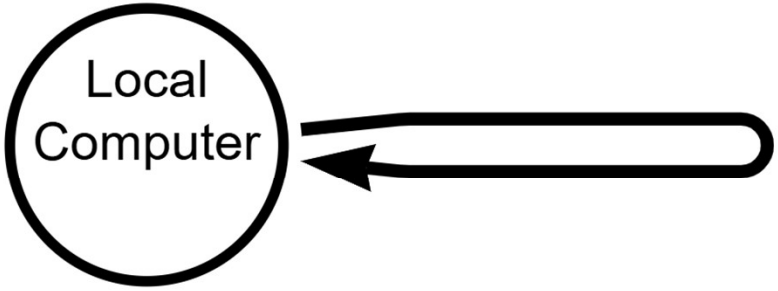
RUN ONCE PHASES

LDAP



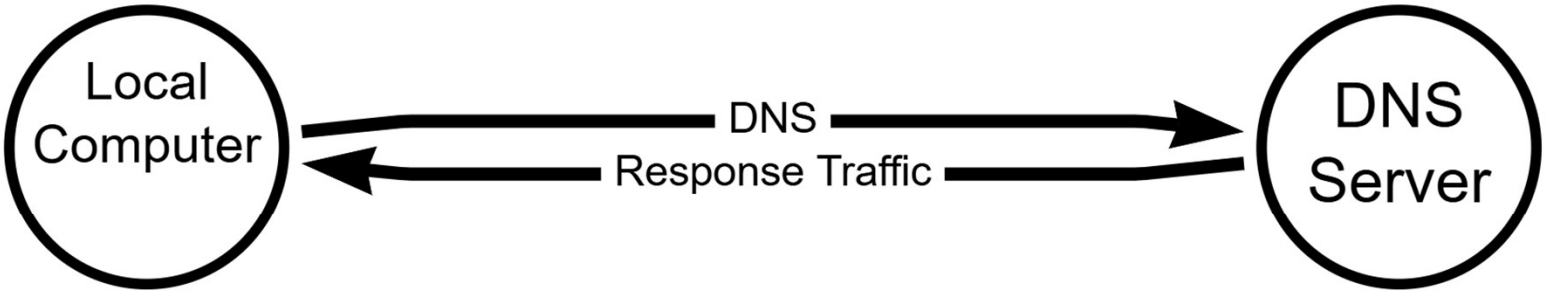
- Reads from **System Management** container in Active Directory
- Discovers:
  - sites, site servers, management points

LOCAL

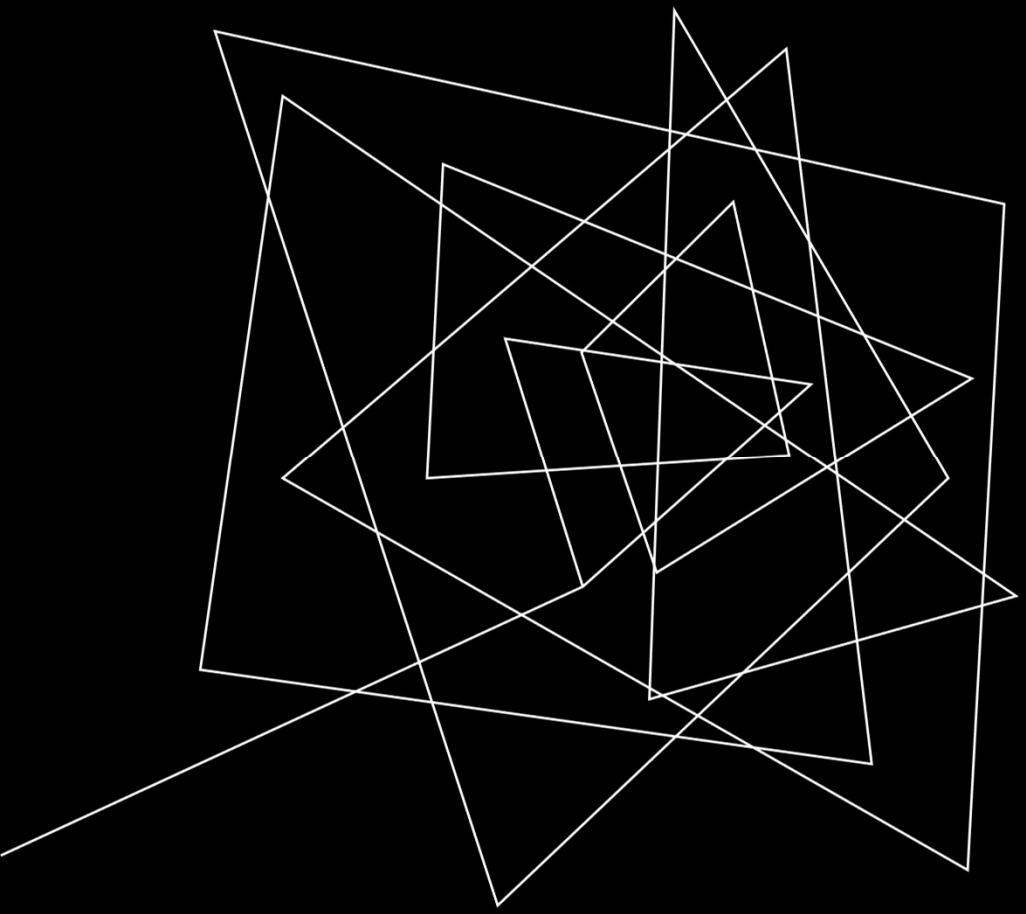


- Reads **client device logs** on disk
- Discovers:
  - management points, distribution points

DNS



- Uses known site codes to query **SRV records**
- Discovers:
  - management points



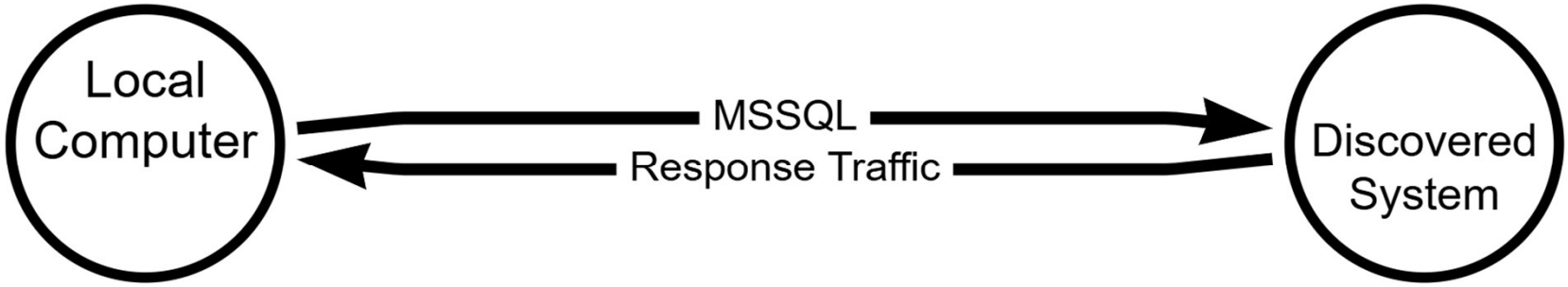
COLLECTION

PER HOST PHASES

REMOTE REGISTRY

- Queries **subkeys accessible to authenticated users**
- Discovers:
  - site servers, site database servers, current user

MSSQL

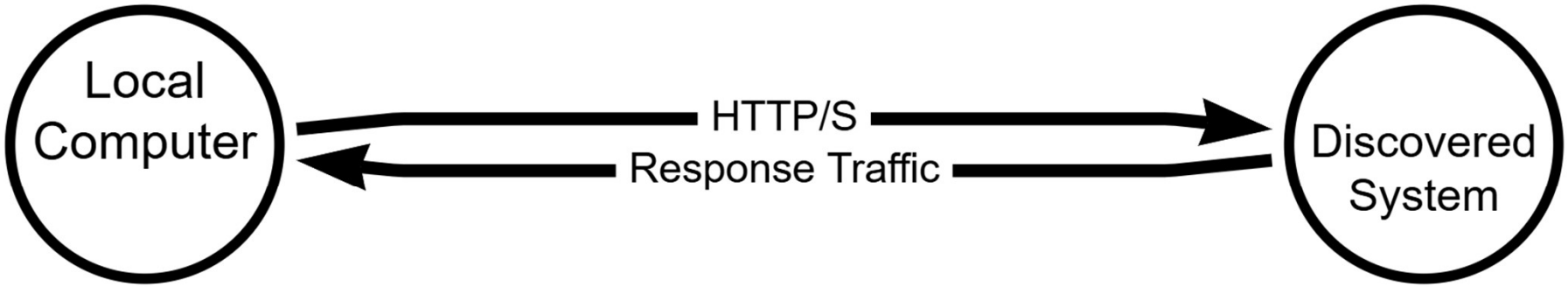


- Identifies **extended protection for authentication** settings
- Discovers:
  - site database servers, server and database principals

ADMINSERVICE

- Queries **AdminService** REST API on SMS Providers
- Discovers:
  - site systems, admin users/roles, client devices, logged in users

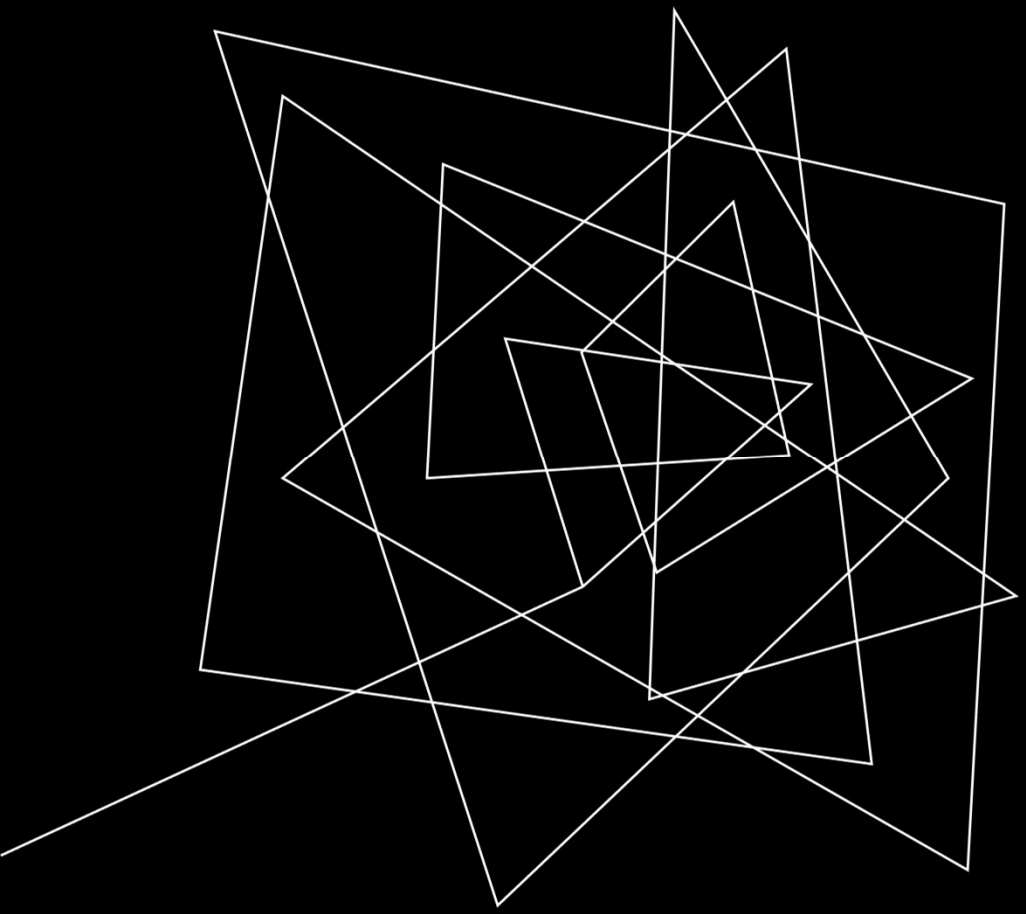
HTTP



- Requests **known HTTP endpoints** and evaluates response codes
- Discovers:
  - management points, distribution points, SMS Providers

SMB

- Identifies **SMB signing** requirements
- Reads SMB **share names and descriptions**
- Discovers:
  - site servers, distribution points



POST-PROCESSING

- Sites
- Admin users and roles
- Site system roles
- Relay to MSSQL
- Relay to AdminService
- Relay to SMB

INTERESTING EDGES

- *CoerceAndRelayToAdminService
- *SameHostAs
- SCCM_AdminsReplicatedTo
- SCCM_AllPermissions
- SCCM_AssignAllPermissions
- *SCCM_HasCurrentUser/HasADLastLogon/HasPrimaryUser
- *SCCM_HasNetworkAccessAccount
- *SCCM_HasStoredAccount
- *SCCM_IsMappedTo

WHAT IS MSSQL?

Microsoft SQL Server

MSSQL ATTACK PATHS

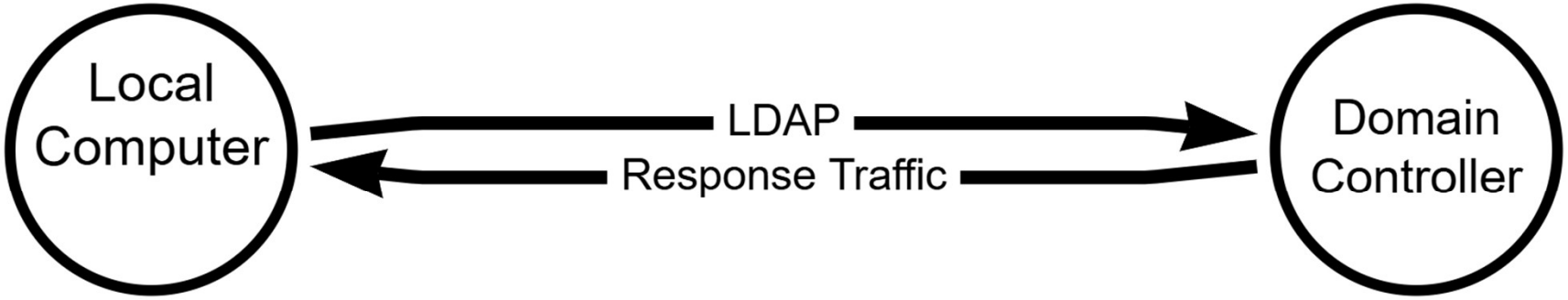
ALLOW ACCESS TO DATABASES... AND MORE

MSSQLHound

- A ~~PowerShell~~ Go OpenGraph collector for MSSQL
- 7 new nodes
- 37 new edges
- Requires an MSSQL server login

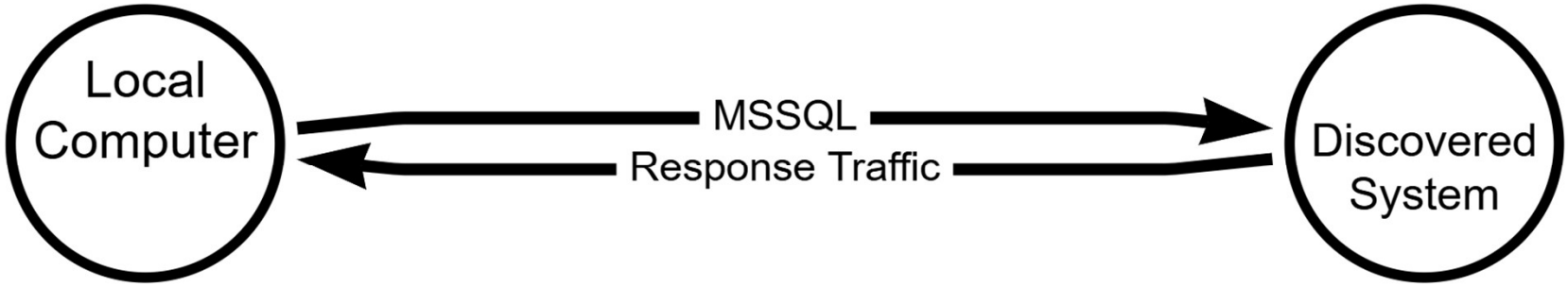
HTTPS://GITHUB.COM/SPECTEROPS/MSSQLHOUND

LDAP

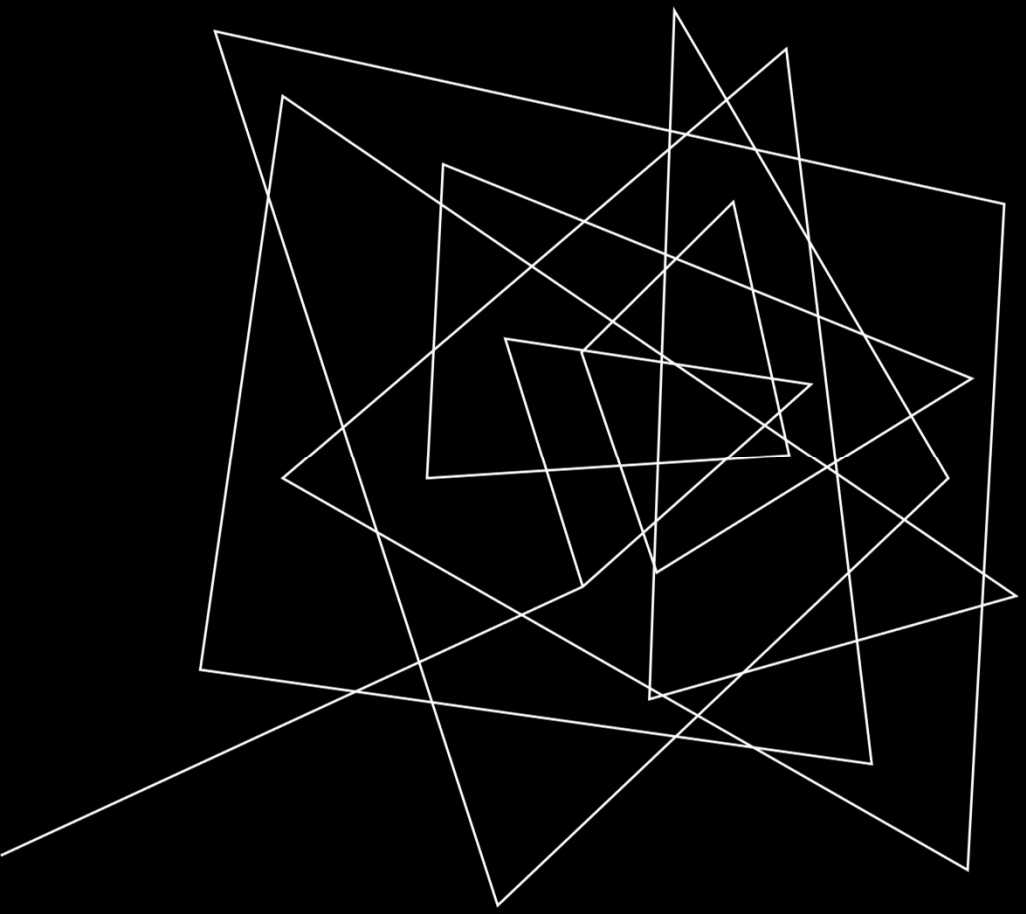


- Searches for computers with MSSQLSvc **service principal names**
- Discovers:
  - MSSQL hostnames, ports, and instance names

MSSQL



- Identifies extended protection for authentication settings
- Discovers:
  - server and database principals, role members, linked servers, stored credentials, and much more



POST-PROCESSING

- Server logins and roles
- Database users and roles
- Members
- Permissions
- Ownership
- Trustworthiness
- Linked servers
- Stored credentials

INTERESTING EDGES

- *CoerceAndRelayToMSSQL
- MSSQL_ChangePassword
- MSSQL_ExecuteAs
- MSSQL_ExecuteAsOwner
- *MSSQL_ExecuteOnHost
- *MSSQL_Get[Admin]TGS
- MSSQL_GrantAny[DB]Permission
- *MSSQL_Has*Cred
- *MSSQL_HostFor
- MSSQL_LinkedAsAdmin

GO PORT

- PR'd and co-authored by 2026 Top Dog Javier Azofra
- **Concurrent** targeting, reducing run duration from 17 minutes to 17 seconds in SCCM Ludus lab environment
- **SOCKS** proxy support
- Cross-platform
- Authenticate with **NT hash** or Kerberos **ticket**
- Direct **upload to BloodHound**
- TDS 8.0 and additional EPA configurations

GRAPHING ATTACK PATHS DEMO

TO DO

- ~~Optimize run duration~~ and Cypher queries
- ConfigManBearPig
  - ~~Port to Python~~
  - Abuse Info
  - DHCP~~/WMI/~~CMPivot collection
  - Add more existing/new attack techniques
- MSSQLHound
  - ~~Port to Go~~
  - Recursive collection from linked databases
  - Database object collection (e.g., stored procedures/tables)

Exposing SCCM and MSSQL Attack Paths in Hardened Environments with OpenGraph

Thank you!

- **Chris Thompson**
- X: @_Mayyhem
- Slack: @Mayyhem
- GitHub: Mayyhem
- LinkedIn: x24